New GDP series for interwar and Soviet Estonia: a Finnish miracle that never was

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Abstract

This paper constructs the first internationally comparable annual series of Estonian GDP per capita for 1919–95 and splices it to existing estimates to give an unbroken series for 1913 and 1919–2022 in 1990 and 2011 international dollars, ready for the Maddison Project Database. The reconstruction proceeds in three steps: a new 1938 benchmark based on a bilateral price comparison with Sweden; a recalculated and extended production-side series for 1919–40; and a new series for the Estonian SSR for 1937–95, built from physical output and employment series drawn from Soviet yearbooks and the Estonian archives and anchored to Bolotin's 1990 benchmark. Independently constructed series agree within 4 per cent where they overlap in 1937–40, and within 1 per cent at the 1991 splice. Estonia grew by 2.8 per cent a year in the 1920s and 2.7 per cent in the 1930s, by 3.9 per cent a

year in 1950–68 and by 0.9 per cent in 1968–89; the transition cut income per head by a third. Estonia stood at 96 per cent of Finnish income per head in 1919 but at only 75 per cent in 1938: almost half of the gap that separated the two countries in 1989 had opened before Soviet occupation. Finland is a revealing comparator for Estonia, but not a measure of what Estonia lost.

JEL codes: N14, O47, E01, P27

Keywords: GDP, historical national accounts, Estonia, Finland, Soviet economy, Maddison Project Database

INTRODUCTION

For Estonians, Finland has long served as an 'alter ego': the neighbour across the Gulf that shared their language, their Lutheran faith and their nineteenth-century awakening, and that showed, through the television signals picked up in Tallinn from the 1960s, what Estonia might have become had it not spent half a century inside the Soviet Union.[1] The comparison is more than nostalgia. It underpins the belief, widespread in the Baltic states, that living standards before the Second World War were comparable to those of the Finns, so that the whole of the gap between the two countries at the restoration of independence in 1991 can be charged to Soviet rule.[2] In its strongest form, found in the Estonian government's 'White Book' on the losses inflicted by occupation, the argument becomes a counterfactual accounting exercise: had Estonia developed as Finland did, its GDP per capita in the early 2000s would have been four to five times what it actually was.[3] Others have doubted that the two economies were ever close, and have pointed out that Finland's own post-war growth was exceptional even by Western European standards, so that the divergence tells us as much about Finland as about Estonia.[4]

What has prevented these claims from being tested is the absence of an annual, internationally comparable GDP series for Estonia before 1995. The existing estimates are partial and rest on conversions that are no longer defensible. Valge's pioneering production-side series covers 1923–38 and was converted into international dollars by assuming, on the authority of Colin Clark's rough figures for 1925–34, that Estonian income per head stood at 90 per cent of Finland's.[5] Klesment, Puur and Valge produced a series for 1950–90 from four sectors of material

[1] A. Lepp and M. Pantti, 'Window to the West: memories of watching Finnish television in Estonia during the Soviet period', *VIEW journal of European television history and culture* 1, no. 3 (2013): 83, 86.

[2] Z. Norkus, *Post-Communist Transformations in Baltic Countries: A Restorations Approach in Comparative Historical Sociology* (Cham: Springer Nature, 2023), 154; J. Å. Dellenbrant, 'Estonia's economic development 1940–1990 in comparison with Finland's', in *Market socialism or the restoration of capitalism?*, ed. A. Åslund (Cambridge: Cambridge University Press, 1992), 206; A. Bocharnikova, 'Economic well-being under plan versus market: the case of Estonia and Finland', *Cato journal* 41, no. 1 (2021): 81–2.

[3] K. Kukk, 'Economic damage', in *The white book: losses inflicted on the Estonian nation by occupation regimes, 1940–1991*, ed. V. Salo et al. (Tallinn: Estonian Encyclopaedia Publishers, 2005), 150; M. Klesment, 'The Estonian economy under Soviet rule: a historiographic overview', *Journal of Baltic studies* 40, no. 2 (2009): 260.

[4] Z. Norkus, 'A comparison of the economic growth of the Baltic states between the two world wars', *World political science* 12, no. 1 (2016): 18; A. M. Kõll, 'The development gap: Estonian adaptation to trade with Western Europe', in *Emancipation and interdependence: the Baltic states as new entities in the international economy, 1918–1940*, ed. A. Johansson, Studia Baltica Stockholmiensia 13 (Stockholm: Almqvist / Wiksell, 1994), 201.

[5] J. Valge, 'Uue majanduse lätteil. Eesti sisemajanduse kogutoodang aastatel 1923–1938', *Akadeemia*, nos. 10-12 (2003): 2725–6; C. Clark, 'Internationaler Vergleich der Volkseinkommen', *Weltwirtschaftliches Archiv* (Tübin-

production, scaled to a 1990 level obtained by deflating a purchasing-power-parity estimate for 2005, of undocumented derivation, published by the UN Economic Commission for Europe (UNECE).[6] The Maddison Project Database (MPD) fills the gap between 1973 and 1990 by backcasting from modern benchmarks with Soviet net material product (NMP) growth rates, a procedure that Maddison himself regarded as inadmissible for the Soviet Union as a whole, and that places Estonian income in 1989 almost a third above the figure reported here.[7] The only estimates that are methodologically consistent with the Maddison tradition are the indirect approximations of Norkus and Markevičiūtė for four benchmark years, 1913, 1922, 1929 and 1938.[8] Between those years, and for the whole of the Soviet period, there is nothing to which a Finnish, Swedish or Soviet series can be compared.

This paper fills that gap. It makes three contributions. First, it constructs a new benchmark estimate of Estonian GDP per capita for 1938, replacing Valge's Finnish bridge with a direct bilateral comparison of consumer prices with Sweden of the kind already used for Lithuania in 1937 and Latvia in 1935, and it uses the new benchmark to recalculate and extend the production-side series to every year from 1919 to 1940.[9] Second, and principally, it provides the first annual series for the Estonian Soviet Socialist Republic (ESSR) covering 1937–95, built from 101 physical product series, agricultural output, transport volumes and employment records drawn from Soviet statistical yearbooks and the Estonian National Archives, and anchored to Bolotin's purchasing-power-parity benchmark for 1990. Third, it splices the two new series to each other, to the 1913 benchmark of Norkus and Markevičiūtė, and to adjusted post-1990 MPD data, to give a continuous series for 1913 and 1919–2022 in both 1990 and 2011 international

gen) 47 (1938): 60.

[6] M. Klesment et al., 'Childbearing and macro-economic trends in Estonia in the XX century', *Rahvastiku-uuringud* (Tallinn) Seeria B, no. 63 (2010): 39–44.

[7] A. Maddison, 'Measuring the Performance of a Communist Command Economy: An Assessment of the CIA Estimates for the USSR', *The Review of income and wealth* (Oxford) 44, no. 3 (1998): 307–8; J. Bolt and J. L. van Zanden, *Maddison Project Database, version 2020*, Groningen Growth and Development Centre, 2020, as quoted in Norkus, *Post-Communist Transformations*, 155. See Table 1.

[8] Z. Norkus and J. Markevičiūtė, 'New estimation of the gross domestic product in Baltic countries in 1913–1938', *Cliometrica* 15, no. 3 (September 2021): 565–674.

[9] A. Klimantas and A. Zirgulis, 'A new estimate of Lithuanian GDP for 1937: How does interwar Lithuania compare?', *Cliometrica* 14, no. 2 (May 2020): 227–281; Z. Norkus et al., 'Benchmarking Latvia's economy: a new estimate of gross domestic product in the 1930s', *Cliometrica* 18 (2024): 251–325; A. Klimantas, 'Lithuanian economy, 1919–1940: stagnant but resilient. The first inter-war GDP time-series estimates and their implications', *The Scandinavian economic history review* 72, no. 2 (2024): 129–158; A. Klimantas et al., 'Reinventing perished "Belgium of the East": new estimates of GDP for inter-war Latvia (1920–1939)', *Cliometrica* (Berlin) 18, no. 3 (2024): 765–835.

dollars, together with a consistent population series. Because the interwar and Soviet-period series are built from different sources and anchored to benchmarks fifty-two years apart, their agreement where they overlap, within 4 per cent in each of the years 1937–40 and within 1 per cent at the 1991 splice, is the strongest available check on the whole reconstruction.

The new series alters the received picture in three respects. First, Estonia's interwar growth was real but unremarkable: income per head rose by 2.8 per cent a year in the 1920s and 2.7 per cent a year in the 1930s, against 5.1 and 3.1 per cent in Finland. Estonia emerged from the war of independence in 1919 at 96 per cent of Finnish income per head, close to the near-parity of 1913, but stood at only 75 per cent in 1938. Measured in logarithms, almost half of the gap between the two countries in 1989 had opened before a Soviet soldier crossed the border. Second, the Soviet period divides sharply in two. Between 1950 and 1968 Estonian income per head grew by 3.9 per cent a year, the fastest sustained growth in the country's history before 1995 and faster than in Finland or the Western European core (hereafter 'Western core'), though not than the Western periphery (Table 4); between 1968 and 1989 it grew by 0.9 per cent a year, and the ground gained was lost again. Estonia stood at 70 per cent of the Western core in 1968 and at 52 per cent in 1989, almost exactly where it had been in 1913. Within the Soviet Union, by contrast, Estonia's position barely changed, from 129 per cent of the Union average in 1940 to 126 per cent in 1989, one of the richest republics throughout. Third, the transition shock was severe: income per head fell by 34 per cent between 1989 and the trough of 1993, and did not regain its 1989 level until 2001.

The paper also speaks to a wider literature. The economic history of the Soviet Union has been rewritten over the past two decades from archival sources, but that work has concentrated on Russia; the non-Russian republics, which held half the Union's population, remain almost entirely without modern national accounts.[10] The Estonian series is built with the same methods as the recent series for Lithuania, so that the two can be compared directly and the procedure

[10]Ekaterina Zhuravskaya et al., 'New Russian Economic History', *Journal of Economic Literature* 62, no. 1 (2024): 47–114; T. Vonyó and A. Markevich, 'Economic growth and structural developments, 1945–1989', ed. M. Morys, The economic history of Central, East and South-East Europe: 1800 to the present (Abingdon: Routledge, 2021), 277–302; I. B. Voskoboynikov, 'Accounting for growth in the USSR and Russia, 1950–2012', *Journal of economic surveys* 35, no. 3 (2021): 870–894; A. M. Kalinovsky, 'Numbers in space: measuring living standards and regional inequality in the Soviet Union', in *Perspectives on the history of global development*, ed. C. R. Unger et al. (Berlin: De Gruyter, 2022), 155–182.

repeated for the remaining republics,[11] and it adds a small, open economy with a strong pre-war statistical tradition to the European peripheral economies for which annual interwar accounts now exist.[12]

Section I sets the scene and reviews the existing estimates; Section II describes sources and method; Section III presents the series; Section IV validates it and assesses its biases; Section V reinterprets Estonian growth under capitalism, planning and transition; Section VI returns to Finland; Section VII concludes.

I | HISTORICAL BACKGROUND AND PREVIOUS ESTIMATES

The territory of present-day Estonia passed from Sweden to Russia in 1710–21 and remained within the Russian Empire until 1918. Two inheritances from the Swedish and Baltic-German centuries distinguished it from the rest of the empire. Lutheran parish schooling made the Estonian and Latvian governorates the most literate in the Tsar's dominions, and the early emancipation of the serfs in 1816–19, followed from the 1860s by the sale of farmsteads to Estonian peasants, created a class of proprietors with an orientation towards market agriculture well before 1914.[13] By the eve of the First World War Tallinn and Narva had become significant manufacturing centres serving the imperial market, so that Estonia was industrialised out of proportion to its size.[14] The best available estimate places Estonian income per head in 1913 at 2,058 Int$$_{1990}$$, within a few per cent of Finland's and about half the level of the Western core (Table 8).[15]

The war, the German occupation of 1918 and the war of independence of 1918–20 reversed these gains. Factories were evacuated eastwards, the urban population fell sharply and the Russian market, which had absorbed the bulk of pre-war Baltic manufactures, was lost for good.[16]

[11] A. Klimantas, 'Accounting for a Soviet republic: Lithuanian GDP time-series, 1940–1995' (Unpublished manuscript, University of Oxford, 2026).

[12] S. Kenny, 'Irish GDP since Independence', *The Economic history review*, 2024, 877–80; S. N. Broadberry and M. Malinowski, 'Living standards in the very long run', ed. M. Morys, The economic history of Central, East and South-East Europe: 1800 to the present (Abingdon: Routledge, 2021), 1–24.

[13] A. Kasekamp, *A history of the Baltic states*, Second edition (London: Palgrave, 2017), 4–5; J. Eellend, 'The harvest of modernization: the formation of agrarianism in Estonia prior to World War I', *Jahrbuch für Geschichte des ländlichen Raumes* 7 (2010): 149–66.

[14] J. Kahk and E. Tarvel, eds., *An economic history of the Baltic countries*, Acta Universitatis Stockholmiensis. Studia Baltica Stockholmiensia, 20 (Stockholm: Almqvist & Wiksell, 1997), 102.

[15] Norkus and Markevičiūtė, 'New estimation', 603.

[16] Kahk and Tarvel, *An economic*, 107; Kõll, 'The development gap', 193–210.

The new republic was therefore built on the land. The land reform of 1919, among the most radical in Europe, broke up the estates of the Baltic-German nobility and created more than 50,000 new smallholdings; co-operative dairies and export associations reoriented agriculture towards butter, bacon and eggs for the British and German markets, in conscious imitation of Denmark.[17] Agriculture still employed some 63 per cent of the labour force in 1934 (Table 11).[18] The Depression brought a banking crisis in 1931–2 and a devaluation in 1933 that pegged the kroon to sterling, and the authoritarian government installed in 1934 presided over a state-led drive into import-substituting industry now described as 'economic nationalism'.[19] Output was still expanding when the Molotov–Ribbentrop pact assigned Estonia to the Soviet sphere in August 1939 and Soviet forces occupied the country in June 1940.

Soviet rule abolished the interwar economic order within a year. Banks, industry and urban property were nationalised in 1940–1; the German occupation of 1941–4 and the renewed fighting of 1944 followed; and in March 1949 the deportation of some 20,000 people broke the resistance of the countryside to collectivisation, which was essentially complete within months.[20] Industry was rebuilt as a specialised component of the all-Union economy: oil shale and electricity in the north-east, cotton textiles in Narva and Tallinn, and later electrical engineering, with the labour supplied by immigration from Russia on a scale that reduced Estonians to little more than three-fifths of the population by 1989.[21] Estonia was consistently among the two or three

[17]H. Jörgensen, 'The inter-war land reforms in Estonia, Finland and Bulgaria: a comparative study', *The Scandinavian economic history review* 54, no. 1 (2006): 64–97; Z. Norkus et al., 'Denmarkisation and agricultural growth in the Baltic countries during the interwar period', *The Scandinavian economic history review*, 2025, 1–20.

[18]M. Morys, 'Economic growth and structural change in Central, East and South-East Europe, 1918–1939', ed. M. Morys, The economic history of Central, East and South-East Europe: 1800 to the present (Abingdon: Routledge, 2021), 181.

[19]B. Bernanke and H. James, 'The Gold Standard, Deflation, and Financial Crisis in the Great Depression: An International Comparison', Essays on the Great Depression (Princeton: Princeton University Press, 2000), 74; A. M. Kõll, 'Economic nationalism: a strategy for late-comers?', in *Economic nationalism and industrial growth: state and industry in Estonia 1934–39*, by A. M. Kõll and J. Valge, Studia Baltica Stockholmiensia 19 (Stockholm: Almqvist / Wiksell, 1998), 11–29; M. Klesment, 'The origins of economic nationalism in interwar Estonia', *Acta historica Tallinnensia* 9 (2005): 118–35; D. Feest, 'Estonia: 'Is this the Estonia we wanted?' How the interwar slump reshaped Estonia's society', in *The Great Depression in Eastern Europe*, ed. K. Richter et al. (Budapest: Central European University Press, 2025), 157–78.

[20]R. Taagepera, 'Soviet Collectivization Of Estonian Agriculture: The Taxation Phase', *Journal of Baltic studies* (Brooklyn) 10, no. 3 (1979): 265; L. Õispuu, *Name list of persons deported from Estonia 1945–1953* (Tallinn: Estonian Repressed Persons Records Bureau, 2014), 12; D. Feest, 'The collectivization of agriculture in the Baltic Soviet republics, 1944–1953', in *The collectivization of agriculture in communist Eastern Europe: comparison and entanglements*, ed. C. Iordachi and A. Bauerkämper (Budapest: Central European University Press, 2014), 100.

[21]T. Parming and E. Järvesoo, *A Case study of a Soviet Republic : the Estonian SSR* (Boulder: Westview Press, 1978), 158–217; P. Ahde and T. Rajasalu, *On the Economic Structure of Estonia and Finland before the 1990's*, Discussion Papers 422 (Helsinki: The Research Institute of the Finnish Economy (ETLA), 1992), 47.

richest republics of the Union on every Soviet measure of income and consumption.[22] Independence was restored in August 1991; a currency board followed in June 1992, prices were freed and enterprises privatised, and the collapse of eastern markets produced the deepest contraction in the country's recorded history before growth resumed in 1995.[23]

The quantitative record of this history is thin. Estonia published no national income accounts before 1940, and the Soviet statistical system measured net material product, which excludes most services and values output at administered prices. Three long series have been attempted, besides the benchmark approximations of Norkus and Markevičiūtė, and their results are compared with the new series in Table 1.

Table 1: The new series compared with previous estimates of Estonian GDP per capita

	This paper	Valge (2003)		Klesment et al. (2010)		MPD (2020)	
Year	(Int$$_{1990}$$)	Level	Diff. %	Level	Diff. %	Level	Diff. %
1923	1,913	1,811	−5.3	—	—	—	—
1924	1,895	2,337	+23.4	—	—	—	—
1929	2,113	2,182	+3.3	—	—	—	—
1932	2,105	2,110	+0.2	—	—	—	—
1938	2,685	2,745	+2.2	—	—	—	—
1950	3,667	—	—	1,676	−54.3	—	—
1960	5,319	—	—	3,949	−25.8	—	—
1970	7,688	—	—	5,377	−30.1	—	—
1973	8,080	—	—	5,621	−30.4	8,657	+7.1
1980	8,785	—	—	5,834	−33.6	—	—
1989	8,958	—	—	6,692	−25.3	11,780	+31.5
1990	8,425	—	—	6,444	−23.5	—	—
1995	6,277	—	—	5,084	−19.0	—	—

Sources: this paper: see online appendix. Valge: J. Valge, 'Uue majanduse lätteil. Eesti sisemajanduse kogutoodang aastatel 1923–1938', *Akadeemia*, nos. 10-12 (2003): 2726, Table 50, Geary–Khamis dollars obtained by scaling Finnish income with Colin Clark's 1925–34 ratio. Klesment et al.: M. Klesment et al., 'Childbearing and macro-economic trends in Estonia in the XX century', *Rahvastiku-uuringud* (Tallinn) Seeria B, no. 63 (2010): 44, Table 2.2, UNECE 2005 purchasing-power parities deflated to 1990 US dollars with the consumer price index, four sectors only. MPD: J. Bolt and J. L. van Zanden, *Maddison Project Database, version 2020*, Groningen Growth and Development Centre, 2020, as quoted in Z. Norkus, *Post-Communist Transformations in Baltic Countries: A Restorations Approach in Comparative Historical Sociology* (Cham: Springer Nature, 2023), 155, converted from Int$$_{2011}$$ at 1.594. *Notes:* Diff. = previous estimate relative to this paper, per cent. '—': not available.

[22] G. E. Schroeder, 'Regional differences in incomes and levels of living in the USSR', in *The Soviet economy in regional perspective*, ed. V. N. Bandera and Z. L. Melnyk (New York: Praeger, 1973), 169; A. Bohnet and N. Penkaitis, 'A comparison of living standards and consumption patterns between the RSFSR and the Baltic republics', *Journal of Baltic studies* (Brooklyn) 19, no. 1 (1988): 35–6.

[23] World Bank, *Estonia: the transition to a market economy*, A World Bank country study (Washington, DC: World Bank, 1993), 1–8; I. Korhonen, 'Progress in Economic Transition in the Baltic States', *Post-Soviet geography and economics* (Palm Beach) 42, no. 6 (2001): 443–4.

The foundation for the interwar period is Valge's production-side estimate of GDP at current and constant prices for 1923–38.[24] Its coverage of agriculture, industry, construction and housing is retained here, but two features limit its use for international comparison. Valge deflated current-price GDP with the cost-of-living index, so that the rapid fall in food prices after 1923 produces a rise of 29 per cent in real income per head in a single year, 1924, when crop, livestock, industrial and transport output each moved by less than 5 per cent.[25] And he expressed the result in international dollars indirectly, by taking Colin Clark's estimate that Estonian output per head averaged 90 per cent of Finland's in 1925–34 and applying that ratio to Maddison's Finnish figures.[26] The resulting level is not far from the one reported here for the 1930s (Table 1), but only because a domestic-currency GDP that is too low is offset by a conversion ratio that is too high. Norkus and Markevičiūtė have since produced indirect, income-side approximations for 1913, 1922, 1929 and 1938 that are consistent with the MPD; their 1913 figure is used here as the pre-war anchor, and their 1938 figure, about 5 per cent below the new benchmark, is an independent check on it.[27]

For the Soviet period the only long series is that of Klesment, Puur and Valge, who extrapolated a 1990 level derived from UNECE purchasing-power parities backwards with an official rouble GDP index for 1980–9 and with a composite index of physical output in agriculture, industry, construction and transport for 1950–79.[28] The four sectors cover only about two-thirds of GDP, since trade and the non-material sphere are omitted, and the UNECE-based level understates income: even in 1995 the series stands 19 per cent below the estimate obtained here from the MPD's 2011 benchmark. Its growth rate of income per head for 1950–78, 4.4 per cent a year, is well above that of the present series, for reasons discussed in Section III. The MPD itself reports Estonian GDP per capita for 1973 and annually from 1980. Its 1990 level descends from Maddison's application of Bolotin's republic-to-Union ratios, taken for 1991 rather than 1990, to a Soviet total itself derived from a bilateral Austria–USSR comparison; its earlier years

[24] Valge, 'Uue majanduse'.

[25] Valge, 2718, 2721; Klesment's re-deflation with wholesale price indices reduces the jump to 21 per cent (M. Klesment, 'Eesti majandusarengu dünaamika näitajaid sõdadevahelisel perioodil', *Tuna* (Tallinn) 38, no. 1 (2008): 29–31; Klesment et al., 'Childbearing', 37–8).

[26] Valge, 'Uue majanduse', 2725; Clark, 'Internationaler Vergleich', 60.

[27] Norkus and Markevičiūtė, 'New estimation'; Norkus, *Post-Communist Transformations*, 154.

[28] Klesment et al., 'Childbearing', 39–44.

are extrapolated backwards with NMP growth rates adjusted by the Union-wide ratio of GDP to NMP, a short cut that the literature on material-product accounting had warned against because the ratio varies across countries and over time, and that overstates growth in the final Soviet decades. The MPD figure for 1989 is 32 per cent above the one reported here.[29]

II | SOURCES AND METHOD

The production approach and the reliability of Soviet statistics

For Estonia only the output side of national income is feasible in either period: consistent annual data on expenditure or factor incomes do not exist for the interwar republic, whose statistical office never compiled national accounts, and for the Soviet period the monetary aggregates that an expenditure or income estimate would require are precisely the figures that cannot be trusted. The series presented here are therefore constructed, in both periods, by extrapolating a benchmark estimate of GDP with constant-price indices of sectoral value-added, each sector weighted by its share of GDP in the benchmark year:

$$GDP_t = GDP_{\text{bench}} \times \sum_{i=1}^{n} w_i \, I_{it},$$

where w_i is sector i's share of benchmark GDP and I_{it} its volume index in year t. The procedure is standard where expenditure and income data are sparse, and is the one used for Ireland, the European periphery and, in the companion studies, Lithuania and Latvia.[30]

For the Soviet period this approach also has an intellectual pedigree that bears directly on the reliability of the result. Soviet practice measured net material product at prevailing prices, excluded most services, double-counted intermediate transactions in its gross-output measures and, above all, understated inflation, so that official constant-price growth was too high: in

[29] A. Maddison, *The World Economy: Volume 1: A Millennial Perspective and Volume 2: Historical Statistics* (Paris: OECD Publishing, 2006), 181–2; Maddison, 'Measuring the Performance', 307–8, 318; J. Češka, 'Comparison of Gross Domestic Product and Net Material Product', ed. W. Krelle, The Future of the World Economy: Economic Growth and Structural Change (New York: Springer-Verlag, 1989), 593; R. Campbell, 'The conversion of national income data of the U.S.S.R. to concepts of the system of national accounts in dollars and estimation of growth rate' (Washington, D.C.), Staff working papers, no. SWP 777 (1985); Vonyó and Markevich, 'Economic growth and structural', 279–80; Table 1.

[30] Kenny, 'Irish GDP', 880–3; Broadberry and Malinowski, 'Living standards', 22; Klimantas, 'Lithuanian economy'; Klimantas et al., 'Reinventing perished'.

Maddison's verdict, the official series 'overstated growth, and understated inflation'.[31] The Western response, begun by Bergson in the 1950s and institutionalised by the CIA from the 1960s, was to rebuild Soviet GNP from the bottom up, repricing output at factor cost, imputing the missing services and, crucially, replacing the official value indices with indices built from physical output series weighted by value-added.[32] The CIA estimates were attacked from both sides. Navrozov and most other critics held that the series overstated Soviet growth by disregarding the deteriorating quality of Soviet products, the over-reporting of output and hidden inflation, and even the physical series were faulted as 'open' statistics designed for external consumption rather than the closed data on which the planners themselves worked; a minority, mostly American, contended that growth was understated because military output, housing and other services were inadequately measured and because physical indicators could not capture technological advance.[33] Khanin, working from Soviet materials in the 1980s, obtained much lower growth rates for the Stalin period but, as Harrison showed, results for 1950–87 that coincided with the CIA's at 3.8 per cent a year.[34] The CIA's own indices were reworked at 1982 prices for the Joint Economic Committee in 1990, the form in which they entered the later literature and the MPD.[35] The assessments by Becker and by Schroeder, the review by Maddison and the recent survey by Vonyó and Markevich all reach the same conclusion: the CIA series contain errors in both directions, most of the criticism 'was not well founded', and the physical-output method remains the best available basis for measuring Soviet growth.[36] Ponomarenko's archival reconstruction of Russian national accounts, the fullest attempt so far to recast Soviet

[31]Maddison, 'Measuring the Performance', 307.

[32]A. Bergson, 'Soviet National Income and Product in 1937', *The Quarterly journal of economics* 64, no. 2 (1950): 208–241; A. Bergson, *The real national income of Soviet Russia since 1928*, Rand Corporation research study ; R-367 (Cambridge, Mass: Harvard University Press, 1961); Joint Economic Committee, *USSR: Measures of Economic Growth and Development, 1950-1980* (Washington, D.C.: Congress of the United States, December 1982), 48; A. Maddison, *Monitoring the world economy 1820-1992*, Development Centre studies (Paris: Organisation for Economic Co-operation / Development, 1995), 328.

[33]L. Navrozov, 'Assessing the CIA's 'Soviet economic indices'', in *Soviet Economy Brink of Reform: Essays in Honor of Alec Nove* (1988), 112–152; A. C. Becker, 'Intelligence Fiasco or Reasoned Accounting?: CIA Estimates of Soviet GNP', *Post-Soviet affairs* (Columbia) 10, no. 4 (1994): 291–329; G. E. Schroeder, 'Reflections on Economic Sovietology', *Post-Soviet affairs* (Abingdon) 11, no. 3 (1995): 197–234.

[34]Navrozov, 'Assessing the CIA's'; M. Harrison, 'Soviet economic growth since 1928: the alternative statistics of G. I. Khanin', *Europe-Asia studies* 45, no. 1 (1993): 144, 159.

[35]Joint Economic Committee, *Measures of Soviet gross national product in 1982 prices* (Washington, D.C.: Congress of the United States, November 1990).

[36]Becker, 'Intelligence Fiasco'; Schroeder, 'Reflections on Economic'; Maddison, 'Measuring the Performance', 309; Vonyó and Markevich, 'Economic growth and structural', 279–80.

data in the System of National Accounts (SNA), reaches the same judgement from inside the Soviet statistical system. The physical output data 'are apparently of high quality and can be used', whereas the official value indices were inflated by the pricing of new products; Khanin's alternative, in Ponomarenko's view, rests on a 'rather primitive methodological base' compared with the CIA's.[37] The literature on the Soviet economy has grown rapidly since the opening of the archives, as Zhuravskaya, Guriev and Markevich document, but its quantitative core for the post-war decades still rests on physical output series of this kind.[38]

It matters, therefore, to be precise about which Soviet data this paper uses and which it avoids. The distortions documented in the literature were concentrated in value aggregates: in the national income and gross output indices at 'comparable prices', in the reporting of hidden price increases as real growth, and in the double counting of the material product system.[39] Khanin's classification of the official data, as Harrison summarises it, is that 'the ease of distortion remained greater for value series than for physical volumes', and that distortion was concentrated in heterogeneous products, above all in machinery and construction, while physical series for homogeneous goods such as grain, milk, electricity, cement or railway freight are comparatively reliable.[40] The sectoral indices below rest almost entirely on physical quantities of this kind, on employment and on inputs; the one place where value data at comparable prices enter, trade turnover, which also feeds the productivity adjustment for finance, is flagged and its bias bounded in Section IV. The claim, then, is not that Soviet statistics were honest, but that the particular statistics used here were recorded for operational planning and had little scope for manipulation. The 'open statistics' objection has correspondingly less force for series that are checked, as these are, against the unpublished returns of the republican statistical administration held in the archives.

[37] Aleksei N. Ponomarenko, *Retrospektyvnye natsional'nye scheta Rossii, 1961–1990* (Moscow: Finansy i statistika, 2002), 16–18, 73–4.

[38] Zhuravskaya et al., 'New Russian Economic History', 47–114; Voskoboynikov, 'Accounting for growth'.

[39] Maddison, 'Measuring the Performance', 307; T. Vonyó, 'War and socialism: why eastern Europe fell behind between 1950 and 1989', *The Economic history review* 70, no. 1 (2017): 248–274.

[40] Harrison, 'Soviet economic growth', 144, 152.

The benchmarks

The 1938 benchmark. Valge's estimate of current-price value-added for 1938 in agriculture, forestry and fishing (250.1 million kroon), industry (132.1 million), construction (14.0 million) and housing (29.0 million) is accepted without revision. His treatment of the remaining services is not. For all other services Valge assumed uniform value-added per worker and scaled a single figure by the service labour force, obtaining 122.9 million kroon in all, less than the contribution of industry and implausibly low for an economy at Estonia's level of urbanisation.[41] This paper rebuilds service value-added from its income components, as in the Lithuanian and Latvian benchmark studies: compensation of employees is estimated from the labour force and average earnings of each service branch, and the remaining components, operating surplus, depreciation and net production taxes, are added in the proportions observed in the Latvian benchmark of 1935, the closest comparator with a full set of accounts.[42] Commerce yields 91.5 million kroon, transport and communications 32.2 million, public services 99.3 million, of which 21.4 million is defence, and household services 5.5 million: 228.5 million in all. Gross value-added rises to 654 million kroon and GDP at market prices, with 17.2 million of taxes less subsidies, to 671 million, or about 592 kroon per head against Valge's 502, which is likewise GDP at market prices and includes the same 17.2 million of product taxes. The sectoral composition is reported in Table 2.

To express this figure in international dollars, a bilateral comparison of consumer prices with Sweden was carried out for 1938, using prices of some fifty consumer goods and services in both countries, weighted by the structure of household expenditure in each. The Fisher price ratio is 1.79 Swedish kronor per Estonian kroon, against an official exchange rate of 1.06: the kroon bought in Estonia roughly 70 per cent more than its exchange value would have bought in Sweden. As in the Lithuanian and Latvian studies, the comparison covers consumer prices only, so that investment and government purchases are converted at the consumer parity. Applied to Swedish GDP per head of 1,931 kronor, the ratio implies that Estonian GDP per head was 54.8 per cent of Sweden's, and since Sweden stood at 4,901 Int$$_{1990}$$ in 1938, Estonia's 1938

[41] Valge, 'Uue majanduse', 2718.

[42] Klimantas and Zirgulis, 'A new estimate'; Norkus et al., 'Benchmarking Latvia's'.

Table 2: Benchmark GDP by sector: Estonia 1938 and 1990, current prices

	1938		1990	
Sector	m. kroon	%	m. roubles	%
Agriculture, forestry & fishing	250.1	38.3	1,250.0	16.9
Industry	132.1	20.2	2,203.4	29.8
Construction	14.0	2.1	818.2	11.1
Transport & communications	32.2	4.9	535.7	7.3
Trade, finance, housing & other commercial services	120.5	18.4	1,322.3	17.9
Public administration, defence, education & health	99.3	15.2	1,064.4	14.4
Other services	5.5	0.8	190.1	2.6
Gross value-added	653.7	100.0	7,384.1	100.0
Taxes less subsidies on products	17.2		—	
GDP at market prices	670.9		7,384.1	
GDP per capita, national currency	592.4		4,705.7	
GDP per capita, Int$$_{1990}$	*2,685*		*8,425*	

Sources: 1938: Valge's estimates for agriculture, industry, construction and housing, with services re-estimated as described in Section II (see online appendix). 1990: B. Bolotin, 'The Former Soviet Union as Reflected in National Accounts Statistics', in *MEMO 3: In Search of Answers in the Post-Soviet Era*, ed. Steve Hirsch (Washington, D.C.: BNA Books, 1992), 184–6, converted to roubles with Bolotin's sectoral purchasing-power parities; services other than trade and transport split using the 1987 Soviet input–output table (D. Steinberg, *Economies of the former Soviet Union: an input-output approach to the 1987 national accounts*, Policy, Research working papers, WPS 1060 (Washington, D.C.: World Bank Group, 1992)). *Notes:* the 1990 figure includes taxes less subsidies within sectoral values ('—': included in sectoral values). Population: mid-year.

benchmark is 2,685 Int$$_{1990}$, or 4,280 Int$$_{2011}$.[43] The result is 2 per cent below Valge's 2,745, but the proximity is accidental: his domestic-currency estimate was too low by 15 per cent and his Finnish bridge too generous by a similar margin. Expressed as a share of Finland, the new benchmark gives 75 per cent for 1938 against Clark's 90 per cent; Norkus and Markevičiūtė's independent approximation gives 71 per cent.[44]

The 1990 benchmark. The Soviet-period series is anchored to Bolotin's estimate of ESSR GDP in 1990, the first systematic attempt to express the accounts of all fifteen republics in international dollars.[45] Bolotin started from the republican GDP totals for 1990 compiled by

[43] J. Bolt and J. L. van Zanden, 'Maddison-style estimates of the evolution of the world economy: a new 2023 update', Maddison Project Database, version 2023, *Journal of economic surveys* 39, no. 2 (2025): 631–671 (hereafter cited as MPD 2023), with Int$$_{2011}$ converted at 1.594 (J. Bolt and J. L. van Zanden, 'The long view on economic growth: New estimates of GDP', How Was Life? Volume II New Perspectives on Well-being and Global Inequality since 1820 (Paris: OECD Publishing, 2021), 12). The price comparison follows the procedure of the Lithuanian and Latvian studies (Klimantas and Zirgulis, 'A new estimate'; Norkus et al., 'Benchmarking Latvia's'); the underlying prices and weights are given in the online appendix.

[44] Norkus, *Post-Communist Transformations*, 154.

[45] B. Bolotin, 'The Former Soviet Union as Reflected in National Accounts Statistics', in *MEMO 3: In Search of Answers in the Post-Soviet Era*, ed. Steve Hirsch (Washington, D.C.: BNA Books, 1992), 181–92.

Goskomstat in 1990–1, 1990 being one of the first years for which republican accounts on an SNA-type definition were compiled and the first with purchasing-power parities, and converted them into dollars with purchasing-power parities estimated, as he put it, 'in practically full conformity' with the International Comparison Project (ICP). The parities, by sector of origin, were 0.40 dollars per rouble in agriculture, 1.80 in industry, 1.10 in construction, 2.80 in transport, 1.45 in trade and 3.30 in other services, or 1.82 for GDP as a whole.[46] For Estonia the result is 8,425 Int$$_{1990}$ per head, the highest of the Baltic republics. Table 2 shows the sectoral composition in roubles, with services other than trade and transport allocated by the 1987 Soviet input–output table;[47] these rouble-price shares, domestic-price weights like the kroon shares of 1938, are the weights with which the volume indices are combined, the dollar conversion applying to the total alone. The shares are at the market prices of 1990, so that product taxes and subsidies are carried within the sectors that bore them; no re-pricing to the adjusted factor cost of Bergson's standard was attempted, since no republican data on the incidence of net product taxes by branch exist. The only socialist economies for which the World Bank reports the ratio of product taxes less subsidies to value-added, Bulgaria and Ukraine in 1980–90, show about 0.5 per cent, and that figure is the bound entered in Table 7 for the distortion of leaving these taxes inside the weights.[48]

Three features of this benchmark need to be stated plainly, because the reconstruction stands or falls with it. First, its level is conservative by the standards of the time. Bolotin's figure for the Soviet Union as a whole, 6,275 dollars per head, is below Maddison's own estimate for 1987, and the 1990 comparison on which it draws, a bilateral Austria–USSR price comparison conducted in the ICP framework, placed the USSR at 32 per cent of the United States where the CIA had placed it at 42 per cent.[49] Second, the parities are Union-wide, so that the conversion assumes that relative prices in Estonia were those of the Soviet Union as a whole. For the material sectors this is a reasonable approximation in a planned economy with administered prices; for services, where Bolotin's parity of 3.30 dollars per rouble reflects the very low rouble wages

[46]Bolotin, 'Former Soviet Union', 181, 184.

[47]D. Steinberg, *Economies of the former Soviet Union: an input-output approach to the 1987 national accounts*, Policy, Research working papers, WPS 1060 (Washington, D.C.: World Bank Group, 1992).

[48]The World Bank, *World Development Indicators*, Accessed on July 9, 2024, Washington, D.C, 2024.

[49]Maddison, 'Measuring the Performance', 318–19.

of the non-market sector, it is less certain. A separate question is the allocation of value-added among services with the 1987 input–output table: the resulting shares of trade and services in Estonian GDP at rouble prices (Table 2) are higher than those implied by the World Bank's 1990 sectoral data for Estonia, 16 per cent for agriculture, 47 per cent for industry and construction and 38 per cent for services.[50] The sensitivity of the series to this allocation is examined in Section IV. Third, there is no alternative. Ponomarenko's retrospective accounts, the most thorough archival reconstruction of Soviet republican national accounts, cover Russia alone, make no dollar conversion and, in their author's words, show that the official exchange rate 'was greatly overvalued', so that any republican benchmark must rest on purchasing-power parities of Bolotin's kind.[51] The other candidates fail on inspection. The World Bank's missions of the early 1990s produced GDP levels for 1980–91 whose publication was discontinued and which no current series reproduces; the UNECE's 'real GDP' indices for 1980–90 were taken directly from NMP growth; the early versions of the Total Economy Database reproduced the UNECE series for the Baltic republics almost exactly; and the long series of Geiger extrapolate the republics from Russian data up to 1990.[52] The CIA's 1982 benchmark cannot be decomposed by republic, and the 1987 input–output table understates services too severely to serve as a benchmark.[53] An independent Estonian benchmark with republic-specific prices is the natural next step; it would alter the level to which the volume indices are anchored and, through the sectoral shares, the weights, but not the indices themselves.

The choice of 1990 rather than an earlier year means that the Soviet-period series is extrapolated backwards for more than fifty years from end-of-period weights. The practical reason is that 1990 is the only year for which republican accounts on an SNA definition and purchasing-power parities exist together. Two further considerations favour it: the reporting of 1989–90, at

[50] The World Bank, *Statistical handbook 1994: states of the former USSR (English)*, 14, Washington, D.C., 1994; World Bank, *Estonia*, 1–8.

[51] Ponomarenko, *Retrospective National Accounts*, 6, 15, 82.

[52] The World Bank, *Estonia - The transition to a market economy (English)*, Washington, D.C., 1993; The World Bank, *World tables 1995 (English)*, Washington, D.C., 1995, 32–3; United Nations Economic Commission for Europe, *Economic survey of Europe in 1994-1995*, New York, 1995, 249; United Nations Economic Commission for Europe, *Economic survey of Europe 1998 no. 1*, New York, 1998, 199; T. Geiger, 'Continuous national gross domestic product (GDP) time series for 195 countries: past observations (1850–2005) harmonized with future projections according to the Shared Socio-economic Pathways (2006–2100)', *Earth system science data* (Katlenburg-Lindau) 10, no. 2 (2018): 847–856.

[53] Steinberg, *Economies of the former Soviet Union*, 66–89.

the height of glasnost, was more transparent than that of any earlier Soviet year, the manipulation of earlier returns having been publicly acknowledged by Baltic statisticians in 1988;[54] and a late benchmark captures the economy at its most diversified, so that branches that barely existed in 1940, electrical engineering, chemicals or oil-shale electricity, receive weights appropriate to their eventual size. The index-number consequences are treated in Section IV; the short answer is that they can be tested directly, because the Soviet-period series meets an independent series with 1938 weights in 1937–40.

Volume indices and their sources

Table 3 summarises the indicators and sources for each sector in each period; the online appendix documents every series year by year. Three general points precede the detail. First, the interwar indices are built from the publications of the Estonian State Statistical Central Bureau, the agricultural surveys of 1925, 1929 and 1939 and the compilations of Valge and of Klesment and Valge, and are weighted by the revised 1938 benchmark.[55] Second, the Soviet-period indices are built from the statistical yearbooks of the ESSR, *Eesti NSV rahvamajandus*, which were published from 1957 and give annual physical output from the mid-1950s. These are supplemented, for the years 1945–55 that the yearbooks cover incompletely or not at all, by the returns of the republican statistical administration held in the Estonian National Archives (fond R-10) and, for the weights within industrial branches, by an archival compilation of 1986 output values for several hundred products (fond R-1969).[56] Archival material is thus used systematically, not merely to fill gaps. Third, the series cover the civilian economy as reported to the republican statistical office. Military production, organised through all-Union enterprises and never reported at the republican level, is not captured: Estonia hosted fewer such plants than Latvia, but the largest, the Dvigatel works in Tallinn and the uranium plant at Sillamäe, lay outside republican statistics, an unquantifiable source of understatement;[57] the size of the Soviet forces stationed in the republic is likewise unknown and is treated as a source of bias in Section IV.

[54] Petras Adlys, 'Statistika: ar viską ji žino? Į klausimus atsako Lietuvos TSR valstybinio statistikos komiteto pirmininko pavaduotojas Petras Adlys', *Tiesa* (Vilnius), no. 194(13813) (August 1988): 2.

[55] Valge, 'Uue majanduse'; M. Klesment and J. Valge, 'Eesti rahvastiku majandustegevuse näitarve XX sajandil', *Rahvastiku-uuringud* (Tallinn) Seeria D, no. 6 (2007).

[56] See Table 3. The archival sources are listed in the online appendix, section A.2.

[57] Kukk, 'Economic damage', 150; Ahde and Rajasalu, *Economic Structure of Estonia*, 66.

Table 3: Volume indicators and sources by sector

Sector	Interwar series, 1919–40 (1938 weights)	Soviet-period series, 1937–95 (1990 weights)	App.
Crops	Physical output of 15 crops weighted by 1938 value-added (Valge; 1939 agricultural survey); yields from M. Klesment and J. Valge, 'Eesti rahvastiku majandustegevuse näitarve XX sajandil', *Rahvastiku-uuringud* (Tallinn) Seeria D, no. 6 (2007)	Physical output; weights from World Bank (1996) and Eurostat agricultural accounts; 1945–65 minor crops interpolated between quinquennial observations	A.1
Livestock	Physical output of meat, milk, eggs and wool; 1938 value-added weights	Physical output; during 1941–4 animal numbers multiplied by pre-war yields	A.1
Forestry and fishing	Timber removals and fish catch; timber for 1919–21 and fish for 1919–27 extrapolated from the crops-and-livestock index	Timber removals and fish catch; 1941–4 interpolated	A.1
Industry	Employment adjusted for real wages (no usable annual output data; Valge's current-price series cannot be deflated by branch)	101 physical product series in 11 branches; branch weights from the 1990 benchmark, product weights from 1986 archival output data (Estonian National Archives)	A.2
Construction	Domestic cement use (production plus net imports), as in Valge; 1919–21 from population	Weighted index of material inputs (1987 Soviet input–output weights), following the CIA procedure	A.3
Transport and communications	Railway passenger- and tonne-kilometres, merchant fleet tonnage, postal items, telegrams and telephone calls	Freight tonne-kilometres and passenger-kilometres for all modes; postal, telephone and broadcasting indicators; 1941–5 interpolated between 1940 and the first post-war observation, with 1943–5 movements from Lithuanian ratios (Section II)	A.4
Trade, finance and insurance	Marketed value-added of agriculture and industry plus deflated imports ('commerce' index); finance assumed to move with commerce	Soviet retail and wholesale turnover at comparable prices (archival and published); finance and insurance from employment with a productivity adjustment	A.5
Housing	Valge's housing value-added for 1929–38, extrapolated with the urban population	Housing stock from construction and population data	A.5
Public administration, defence, health and education	Civil services assumed to move with GDP; defence from military personnel (15,900 in 1921–39)	Employment by branch (Soviet yearbooks, archival records); defence from the 1940 and 1990 totals	A.6
Other services	Total population	Employment	A.6

Sources: Interwar: J. Valge, 'Uue majanduse lätteil. Eesti sisemajanduse kogutoodang aastatel 1923–1938', *Akadeemia*, nos. 10-12 (2003): 2202–2735; M. Klesment and J. Valge, 'Eesti rahvastiku majandustegevuse näitarve XX sajandil', *Rahvastiku-uuringud* (Tallinn) Seeria D, no. 6 (2007); interwar statistical yearbooks and agricultural surveys. Soviet period: *Eesti NSV rahvamajandus* statistical yearbooks (1957–89), *Eesti statistika aastaraamat* (1990–98), Estonian National Archives (ERA, fond R-10 and R-1969), German occupation statistical bulletins (*Statistische Berichte für das Ostland*, 1942–44), and World Bank (1993, 1996). Full source lists by series are given in the online appendix, sections A.1–A.7.

Agriculture, forestry and fishing. In both periods the index combines physical output of crops (fifteen products, including fodder) and of livestock products (meat, milk, eggs and wool) with timber removals and the fish catch, each sub-index weighted by its benchmark-year value-added. Interwar output comes from Klesment and Valge's compilation and the agricultural surveys, Soviet-period output from the yearbooks and, for the early post-war years, the archives, where some minor crops are reported only quinquennially and have been interpolated.[58] The weights for the Soviet period derive from the World Bank's 1990 agricultural accounts for Estonia and, for the split within grains, from the Eurostat economic accounts for agriculture.

Industry. No annual volume index of interwar industrial output exists, and Valge's current-price series cannot be deflated by branch for want of producer price indices. The interwar industrial index is therefore built from industrial employment adjusted for the movement of real wages, a recognised second-best when the share of industry in GDP is modest.[59] For the Soviet period, by contrast, physical output series could be assembled for 101 products in eleven branches: electricity, petroleum products, other fuels including oil shale, metals, chemicals, machinery and metalworking, wood and paper, building materials, light industry, food and a residual. Products are aggregated to branches with the 1986 archival weights (fond R-1969) and branches to industry with the branch structure of the 1990 benchmark study, the petroleum branch being separated from other fuels with the value-added shares of the 1987 input–output table, following the CIA's procedure for Soviet GNP.[60]

Construction. The interwar index follows Valge in tracking domestic consumption of cement, production plus net imports; it is volatile in the 1920s, when timber construction, which the index does not capture, was still important.[61] The Soviet-period index is a weighted index of the material inputs into construction, with weights from the 1987 input–output table, which is the CIA's procedure.[62]

Transport and communications. The interwar index combines railway passenger- and tonne-

[58] Klesment and Valge, 'Eesti rahvastiku'; online appendix, section A.1.

[59] Broadberry and Malinowski, 'Living standards', 22; M. Thomas and C. Feinstein, 'The value-added approach to the measurement of economic growth', ed. T. W Guinnane et al., History Matters: Essays on Economic Growth, Technology, and Demographic Change (Stanford: Stanford University Press, 2003), 427–8.

[60] Joint Economic Committee, *USSR: Measures*, 48; Steinberg, *Economies of the former Soviet Union*, 66–9; A. Maddison, *Historical Statistics of the World Economy: 1-2008 AD*, 2010, 312.

[61] Valge, 'Uue majanduse', Table 35.

[62] Steinberg, *Economies of the former Soviet Union*, 66–89.

kilometres, the tonnage of the merchant fleet, and postal items, telegrams and telephone calls. The Soviet-period transport index combines freight tonne-kilometres and passenger-kilometres across all modes; the communications index combines postal and telegraph traffic, telephone calls and radio and television sets in use.

Trade, finance and housing. For the interwar years a single 'commerce' index tracks the marketed value-added of agriculture and industry together with deflated imports, and finance and insurance are assumed to move with it. Under planning, industrial products circulated by allocation rather than through markets, so the Soviet-period trade index is built instead from the retail and wholesale turnover statistics at comparable prices compiled by the republican statistical office, most of them archival; these are the only value series used in the reconstruction, and the hidden inflation they may contain is bounded in Section IV. Finance and insurance follow employment, adjusted for productivity with the ratio of trade turnover to trade employment, as the CIA did; housing follows the housing stock, derived from construction and population.[63]

Public and other services. Interwar public services other than defence are assumed to move with the rest of the economy, since no consistent series of civil employment exists; defence follows military personnel, which is known for 1919–20 and 1939 and held constant in the intervening years. Soviet-period public administration, health and education follow employment by branch, as in the CIA's accounts, with defence derived from the 1940 and 1990 totals. Other, mainly domestic, services follow population before 1940 and employment thereafter.

The war years. Estonia was under German civil administration from 1941 to 1944, and the *Reichskommissariat Ostland* published statistics, the *Statistischer Jahresbericht* for 1942, the *Statistische Berichte für das Ostland* of 1942–4 and *Ostland in Zahlen*, that cover crops, industrial output and employment but little else; the three changes of regime also destroyed part of the archival record.[64] Where these fail, forestry and fishing, transport, finance, administration and other services are interpolated between 1940 and the first post-war observation, and the movement between 1943 and 1945, for which no Estonian figures could be found, is taken from the ratios observed in Lithuania, where documentation is better; the collapse of the finance index

[63] Maddison, 'Measuring the Performance', 319; Joint Economic Committee, *USSR: Measures*, 48.

[64] J. Dagys, 'Visuomeninio produkto ir nacionalinių pajamų apimties buržuazinėje Lietuvoje klausimu', *Liaudies ūkis*, May 1959, 140.

in 1944 and the doubling of the public-administration index between 1944 and 1945 in Table A3 are artefacts of that borrowed movement meeting the first post-war Estonian employment return. The estimates for 1941–5 are therefore 'rough' in Feinstein's sense, with a margin of 15–25 per cent, and the 1944 trough is an order of magnitude rather than a measurement.[65] Because the Soviet-period series is extrapolated backwards through the war, an error in the wartime linkage would shift its level in 1937–40, so that the overlap tests the linkage together with the rest of the extrapolation; the spliced series, which follows the interwar series to 1940, is affected only through the 1940 average.

Population

Estonia's borders changed once, when the area east of Narva and most of Petseri county, together about 6 per cent of the interwar population, were transferred to the Russian republic in 1944–5. The series to 1944 refer to the borders of the republic and the series from 1945 to present-day borders, which is the convention of the MPD; the transferred districts were rural and poorer than the national average, so that the break in income per head is likely to be small, and no adjustment has been made.[66] The population series rests on the censuses of 1922, 1934, 1959, 1970, 1979 and 1989 and the year-end estimates of Statistics Estonia and the 'Baltic 100' project, with mid-year figures as the average of consecutive year-ends; wartime losses, deportations and the flight of some 70,000 people to the West in 1944 are taken from the demographic literature, and the Soviet garrison of 1940–1 is added to the resident population (Table A1).

Aggregation and splicing

Each of the two new series is first compiled in constant benchmark-year prices and then expressed in Int$$_{1990}$ through its benchmark. The interwar series covers 1919–40 and the Soviet-period series 1937–95, so that the two overlap for four years. They are spliced at 1940 by taking the average of the two estimates for that year, each series otherwise keeping its own level, so that the splice moves the 1940 figure by less than half a per cent; the Soviet-period series is in

[65] C. H. Feinstein, *National income, expenditure and output of the United Kingdom, 1855-1965* (Cambridge: Cambridge University Press, 1972), 21.

[66] Bolt and Zanden, 'The long view', 30.

turn spliced at 1991, in the same way, to a modern series obtained by backcasting the MPD's 2011 benchmark with the World Bank's constant-price growth rates of GDP per capita, which are used in preference to the MPD's own 1990–2010 growth rates for reasons explained in Section IV.[67] The 1913 figure is the benchmark of Norkus and Markevičiūtė, 2,058 Int$$_{1990}$, taken as published; there are no estimates for 1914–18, and the change between 1913 and 1919 therefore combines their income-side approximation with the new production-side estimate and should be read with that caveat. Following Bolt and van Zanden, the series is compiled in Int$$_{1990}$ and converted to Int$$_{2011}$ at 1.594, and both versions are reported.[68]

III | THE NEW SERIES

The annual series of GDP per capita and population for 1913 and 1919–2022 is reported in full in Table A1 of the appendix, and the sectoral volume indices from which it is built in Tables A2 and A3. Table 4 summarises growth by sub-period alongside the comparators that recur in the discussion below, and Table 5 gives the growth of each sector.

Table 4: Average annual growth of GDP per capita by sub-period (per cent)

Period	Estonia	Lithuania	Finland	Nordics	Western core	Western periphery	ECE	USSR
1913–38	1.1	0.9	2.1	1.9	1.0	1.2	0.8	1.7
1919–29	2.8	4.5	5.1	3.3	3.5	3.1	4.8	9.2
1929–38	2.7	3.0	3.1	2.1	1.1	0.8	0.1	5.0
1938–50	2.6	0.9	1.4	2.1	0.6	−0.1	0.8	2.4
1950–68	3.9	4.9	3.6	3.1	3.6	5.2	3.9	3.4
1968–89	0.9	1.8	3.6	2.5	2.4	3.0	1.7	1.5
1938–89	2.4	2.7	3.1	2.6	2.4	3.0	2.3	2.4
1989–95	−5.8	−6.6	−0.8	1.6	1.3	2.1	−1.6	−8.0
1995–2022	4.2	5.0	1.7	2.1	1.5	1.4	4.4	3.9

Sources: Estonia: see online appendix. Lithuania: A. Klimantas, 'Accounting for a Soviet republic: Lithuanian GDP time-series, 1940–1995' (Unpublished manuscript, University of Oxford, 2026). Other series: J. Bolt and J. L. van Zanden, 'Maddison-style estimates of the evolution of the world economy: a new 2023 update', Maddison Project Database, version 2023, *Journal of economic surveys* 39, no. 2 (2025): 631–671 (hereafter cited as MPD 2023), supplemented as documented in the online appendix. *Notes:* compound annual rates between end-point years, Int$$_{1990}$. Nordics: Denmark, Finland, Iceland, Norway and Sweden; Western periphery: Finland, Greece, Iceland, Ireland, Italy, Portugal and Spain; Western core: Austria, Belgium, Denmark, France, Germany (West Germany 1950–97), the Netherlands, Norway, Sweden, Switzerland and the UK; ECE (East-Central Europe): Bulgaria, Czechoslovakia, Hungary, Poland, Romania and Yugoslavia; USSR: Soviet Union within 1990 borders (former Soviet Union after 1991).

[67] MPD 2023; The World Bank, *World Development Indicators 2024*.
[68] Bolt and Zanden, 'The long view', 12, 30.

Table 5: Average annual growth of sectoral value-added volume indices, Estonia (per cent)

Sector	1919–29	1929–38	1938–50	1950–68	1968–89	1989–91	1938 level regained in	Peak year
Crops	2.6	2.8	1.1	2.2	−1.1	−8.7	1947	1968
Livestock	7.7	1.5	−3.9	3.7	2.0	−8.6	1960	1987
Forestry and fishing	4.7	0.6	0.9	7.6	1.4	−12.4	1947	1976
Primary sector	5.2	1.9	−1.3	3.1	0.6	−8.9	1959	1986
Food	—	—	2.2	8.8	2.1	−9.7	1948	1976
Textiles	—	—	4.6	9.0	0.9	−5.5	1949	1987
Other light industries	—	—	2.8	7.1	1.8	−12.4	1948	1988
Heavy industries	—	—	5.1	12.0	1.2	−8.6	1948	1980
Industry	11.5	5.7	5.0	9.5	1.4	−9.1	1948	1988
Construction	11.7	7.6	5.4	9.3	0.9	−9.9	1947	1977
Transport and communications	10.7	2.0	8.0	8.9	4.9	−4.2	1945	1989
Trade, accommodation and catering	—	—	1.2	9.6	4.7	−22.0	1949	1990
Housing	1.8	6.8	1.1	2.8	2.0	−4.6	1949	1986
Public services	−5.3	1.3	4.0	1.2	1.8	−2.5	1945	1984
GDP	3.3	2.9	1.9	5.4	1.7	−8.7	1947	1989

Sources: see online appendix. *Notes:* compound annual growth between end-point years of constant-price value-added indices (1938 = 100). Industrial branch indices are available from 1938 only. '1938 level regained': first post-war year (1945 or later) in which the index is at or above its 1938 value; 'Peak year': year of the highest index value up to 1990. Public services comprise administration, defence, health and education. Growth in 1938–50 is affected by the wartime trough of 1944, for which Estonian sectoral movements are partly extrapolated from Lithuanian ratios (Section IV). The 1989–91 column ends in 1991, the splicing year beyond which the final series follows adjusted Maddison Project data.

On the eve of the First World War Estonian income per head stood at 2,058 Int$$_{1990}$$. The war and the fighting of 1918–20 cut it, on the two benchmarks available, which rest on different methods, by more than a fifth, to 1,600 Int$$_{1990}$$ in 1919, a level that was not regained until 1929. Recovery in the 1920s was steady rather than rapid, 2.8 per cent a year, and was carried by the sectors that had suffered most in the war: industry, construction and transport each grew by more than 10 per cent a year from their very low 1919 base, while agriculture, which had been less damaged, grew by 5.2 per cent a year (Table 5). The Depression was shallow. Income per head fell from 2,282 Int$$_{1990}$$ in 1930 to 2,105 Int$$_{1990}$$ in 1932, a decline of 8 per cent, and had recovered by 1934. The recovery after 1933 was the fastest phase of interwar growth, with income per head rising by more than 4 per cent a year to 1938; over 1929–38 as a whole construction grew by 7.6 per cent a year, industry by 5.7 per cent and housing by 6.8 per cent, and income per head reached 2,685 Int$$_{1990}$$ in 1938 and an interwar peak of 2,771 Int$$_{1990}$$ in 1940, the year of the Soviet occupation. Over the two decades of independence as a whole, income per head grew by 2.8 per cent a year between 1919 and 1938, but by only 1.1 per cent a year measured from 1913: the recovery from the war absorbed most of the gain.

The Second World War reduced income per head to 1,909 Int$$_{1990}$$ in 1944, a third below the level of 1940, though as Section II explained the 1944 figure is partly extrapolated and should not be pressed. Reconstruction was rapid: GDP regained its 1938 level in 1947, industry in 1948 and income per head reached 3,667 Int$$_{1990}$$ in 1950, more than a third above its 1938 level. The result is striking beside Lithuania, where income per head grew by less than 1 per cent a year over 1938–50, and Finland, where it grew by 1.4 per cent (Table 4), and two things account for most of the difference. The first is the denominator: the resident population was 8 per cent smaller in 1950 than in 1938, so that total GDP rose by a quarter where income per head rose by a third. The second is the weight of investment in oil shale, electricity and textiles in the fourth five-year plan, which took industrial output to 1.8 times its 1938 level and transport to 2.5 times by 1950 (Table A2). Because the Soviet-period series is anchored in 1990, the 1950 level does not depend on the wartime observations but on growth over 1950–90. Agriculture did not share in this recovery. Crop output was back at its 1938 level by 1947, but livestock output, which had been halved by the war and was then disrupted by collectivisation, did not regain it until 1960, and the agricultural sector as a whole only in 1959, twenty years after the pre-war peak.

The eighteen years from 1950 to 1968 are the most dynamic in the series before 1995. Income per head grew by 3.9 per cent a year, from 3,667 to 7,348 Int$$_{1990}$$, with total GDP growing by 5.4 per cent a year. Heavy industry grew by 12.0 per cent a year, textiles by 9.0 per cent and food processing by 8.8 per cent; construction, transport and trade each grew by around 9 per cent a year; and agriculture, now recovering, by 3.1 per cent. Growth then slowed sharply and, after 1973, almost ceased. Income per head rose by 0.9 per cent a year between 1968 and 1989, and by only 0.6 per cent a year after 1973; between 1980 and 1989 it rose by 0.2 per cent a year. The sectoral indices show that the slowdown was general. Food processing peaked in 1976, construction in 1977, heavy industry in 1980 and the housing stock in 1986; among the material sectors only textiles, other light industry, livestock, transport and trade continued to grow into the late 1980s, together with the employment-based service indices, and industrial output as a whole peaked in 1988. The series reaches its Soviet-era maximum of 8,958 Int$$_{1990}$$ in 1989.

The collapse that followed was deeper than anything in the interwar record. Income per head fell by 34 per cent between 1989 and 1993, to 5,870 Int$$_{1990}$$, with the sharpest fall, of 19 per

cent, in 1992, the year in which the kroon replaced the rouble and trade with the former Union contracted. Growth resumed in 1995, averaged 4.2 per cent a year to 2022, and carried income per head past its 1989 level in 2001 and to 18,863 Int$$_{1990}$ (30,067 Int$$_{2011}$) in 2022, second among the Baltic states behind Lithuania.

Table 1 compares the new series with the three earlier estimates. For the 1930s the new series and Valge's differ by only a few per cent, which, as explained above, reflects offsetting errors in his domestic-currency total and his conversion ratio rather than agreement in method. For the 1920s the differences are larger and change sign, and the 29 per cent rise that Valge's deflation produces for 1924 becomes a fall of 1 per cent in the new series, in which crops, livestock, industry and transport move by between -2 and $+5$ per cent in that year and the small net fall reflects the volatile cement-based construction index (Table A2). The series of Klesment, Puur and Valge lies a quarter to a third below the new one throughout 1960–89, and more than half below in 1950, because of its low UNECE benchmark and the omission of trade and the non-material sphere; its implied growth of income per head over 1950–78, about 4.4 per cent a year, is well above the 3.0 per cent found here, because their index omits the non-material sphere, whose employment-based components grew far more slowly than material production, and weights the four material branches by their shares of net material product, which overstate the share of industry in GDP. The MPD figures, by contrast, lie above the new series, by 7 per cent in 1973 and by 32 per cent in 1989, because they carry modern benchmarks backwards with NMP growth rates that overstate real growth in the final Soviet decades.

IV | VALIDATION, LIMITATIONS AND SENSITIVITY

The overlaps

The reconstruction can be tested at two points: the interwar and Soviet-period series both cover 1937–40, and the Soviet-period and modern series, the latter backcast from the MPD's 2011 benchmark, both cover 1990–5 (Table 6).

In 1937–40 the two series differ by -1.6, $+3.6$, -3.0 and -0.3 per cent, all within the 5

Table 6: Overlapping estimates of Estonian GDP per capita at the two splicing points

Year	Earlier-benchmark series (Int$$_{1990}$$)	Later-benchmark series (Int$$_{1990}$$)	Difference (%)	Feinstein grade
(a) Interwar series (1938 benchmark) and Soviet-period series (1990 benchmark)				
1937	2,722	2,678	−1.6	Firm
1938	2,685	2,782	+3.6	Firm
1939	2,736	2,654	−3.0	Firm
1940†	2,776	2,767	−0.3	Firm
(b) Soviet-period series (1990 benchmark) and adjusted MPD series (2011 benchmark)				
1990	8,425	8,172	+3.0	Firm
1991†	7,504	7,560	−0.8	Firm
1992	5,314	6,069	−14.2	Good
1993	4,733	5,870	−24.0	Rough
1994	4,799	5,899	−22.9	Rough
1995	4,657	6,277	−34.8	Conjectural

Sources: see online appendix; adjusted MPD series from J. Bolt and J. L. van Zanden, 'Maddison-style estimates of the evolution of the world economy: a new 2023 update', Maddison Project Database, version 2023, *Journal of economic surveys* 39, no. 2 (2025): 631–671 (hereafter cited as MPD 2023) with 1990–2010 growth replaced by World Bank constant-price growth backcast from the 2011 benchmark (The World Bank, *World Development Indicators*, Accessed on July 9, 2024, Washington, D.C, 2024). *Notes:* Difference = (later − earlier)/earlier in panel (a) and (earlier − later)/earlier in panel (b), so that a positive number always means that the 1990-based series is the higher one. Feinstein's grades: firm <5 per cent, good 5–15, rough 15–25, conjectural >25 (C. H. Feinstein, *National income, expenditure and output of the United Kingdom, 1855-1965* (Cambridge: Cambridge University Press, 1972), 21). † Splice year: the final series takes the arithmetic average of the two estimates.

per cent margin that Feinstein reserved for 'firm' estimates, and not of one sign.[69] The test is demanding. The two series share no sources, the one resting on the republic's own statistics and the other on ESSR yearbooks and post-war archives; no weights, one using the structure of 1938 and the other that of 1990; and no benchmark, one a bilateral price comparison with Sweden, the other a purchasing-power-parity study of the Soviet republics fifty-two years later. That the Soviet-period series, extrapolated backwards over half a century of war, collectivisation and forced industrialisation, arrives within 4 per cent of an independently measured level is the strongest evidence available that the net error of the extrapolation is small. It cannot exclude errors that happen to offset, such as an overstatement of growth from hidden inflation in trade turnover against an understatement from the index-number assumptions. But the two index-number biases discussed below, late base-year weights and fixed value-added weights, would both understate growth and so would both place the backcast 1937–40 level above the interwar series; the overlap therefore tests their combined size. A net understatement of growth of as little as 0.1 percentage point a year over 1940–90 would place the Soviet-period series 5 per cent above the interwar series in every overlap year, which is not observed. Nor can the agreement be the product of a low benchmark offsetting understated growth: the 1990 level is, if anything, 3 per cent above the independent 2011-based estimate (Table 6, panel b), which would push the backcast levels up, not down. Broadberry and Burhop regarded time-series projections as consistent with direct benchmarks when they agree within 10 per cent, and judged a thirty-year projection that landed within 4 per cent a success;[70] the Estonian overlap meets that standard over a longer span in each of four consecutive years.

The second overlap is less tidy, and instructively so. In 1990 the Soviet-period series stands 3.0 per cent above the modern series and in 1991 less than 1 per cent below it; thereafter the gap widens to 14 per cent in 1992 and 35 per cent by 1995. The agreement in 1990–1 validates the 1990 benchmark, and in particular the conversion from roubles to dollars: Bolotin's figure rests on purchasing-power parities, not exchange rates, and the modern series derives from an entirely

[69] Feinstein, *National income*, 21.

[70] S. N. Broadberry and C. Burhop, 'Comparative Productivity in British and German Manufacturing Before World War II: Reconciling Direct Benchmark Estimates and Time Series Projections', *The Journal of Economic History* 67, no. 2 (2007): 323–6; S. N. Broadberry and C. Burhop, 'Resolving the Anglo-German Industrial Productivity Puzzle, 1895–1935: A Response to Professor Ritschl', *Journal of Economic History* 68, no. 3 (2008): 933.

independent ICP round, that of 2011, carried backwards with the World Bank's constant-price growth rates, so that two routes that share no data arrive at the same level within 3 per cent. The divergence after 1991 arises because the Soviet-period series continues to value the collapse of Soviet-era output at 1990 weights and, in trade, relies on deflated turnover figures that understate inflation: it records what happened to the Soviet-era economy, not what replaced it. The modern series is therefore preferred from 1991, the year in which the two agree. It is anchored on the MPD's 2011 figure for Estonia, 21,997 Int$$_{2011}$$ or 13,800 Int$$_{1990}$$, and carried back to 1990 with the World Bank's constant-price growth of GDP per head rather than with the MPD's own growth, because the two sources' Estonian growth rates converge only in 2011: the MPD's path implies a rise of 28 per cent between 1990 and 2011 where the national accounts record 69 per cent, so that its 1990 level, 17,247 Int$$_{2011}$$, stands 32 per cent above the World Bank-based 13,033. Since the two share the 2011 anchor, the whole of the difference between the MPD and the new series at the end of the Soviet period lies in the growth attributed to 1990–2010, not in the level; and Bolotin's independent 1990 figure falls within 3 per cent of the World Bank-based one, which is the agreement that Table 6 reports.[71] The MPD's 1989 figure lies 32 per cent above the new series (Table 1), but its implied transition fall, 36 per cent against 34 here, is similar: the two agree on the depth of the collapse though not on the level from which it started.[72] The 1990-weighted and the 2011-based series bracket the true fall. The former values the collapse of the Soviet-era branches at their 1990 weights and misses the new activities, above all in services, that the IMF noted were emerging unrecorded; the latter, like Hamilton and de Vries's recent extrapolation of the post-Soviet republics from mid-1990s benchmarks, carries the weights of the surviving branches backwards and so records a shallower decline.[73]

[71] MPD 2023; The World Bank, *World Development Indicators 2024*; the MPD's 1990 level is discussed in Section I.

[72] The comparison is developed in Klimantas, 'Accounting for a Soviet republic'.

[73] O. Havrylyshyn et al., *Growth Experience in Transition Countries, 1990-98* (Washington DC: International Monetary Fund, 1999), 11; C. Hamilton and G. J. de Vries, 'The structural transformation of transition economies', *World development* 191 (2025): 17; M. De Broeck and V. Koen, *The Great Contractions in Russia, the Baltics and the Other Countries of the Former Soviet Union: A View From the Supply Side.*, IMF Working Papers (Washington, D.C: International Monetary Fund, 2000), 22.

A single late benchmark and the index-number problem

The Soviet-period series extrapolates 1990 weights backwards over 1937–90, a period of profound structural change. An index with a late base year assigns to each sector the relative prices of the end of the period, and since the sectors that grew fastest were, as a rule, those whose relative prices fell, late weights give less importance to fast-growing sectors and so understate growth; conversely, early weights overstate it. Feinstein and Thomas put the point generally: an index with an early base 'will always show a greater increase in quantities than one constructed with a base year late in the period'.[74] This is the effect that Gerschenkron showed for Soviet machinery, and it was large in the Stalin period: Bergson's GNP for 1928–37 grows by 175 per cent at 1928 prices and by 62 per cent at 1937 prices.[75] Three considerations establish that it is far smaller for the series presented here.

First, the effect is concentrated in the early phases of industrialisation, when relative prices change fastest; Jonas and Sardy showed that 'most of the action which leads to the Gerschenkron effect takes place early'.[76] For the post-war Soviet economy the measured effect is modest. When the CIA moved its base year from 1970 to 1982 prices, its estimate of Soviet GNP growth over 1951–84 fell by about half a percentage point a year, from 5.6 to 5.2 per cent for the 1950s and from 3.2 to 2.4 per cent for the 1970s; an earlier comparison of 1960, 1970 and 1976 prices produced differences of a few tenths of a point, which the CIA judged 'not large'.[77] Half a point a year over forty years compounds to roughly a fifth of the level; this indicates the order of magnitude to be expected for the post-war decades rather than a bound, since the Estonian extrapolation also spans the forced industrialisation of the 1940s and 1950s, in which the effect is largest. Second, the bound must therefore come from the data, and the series can be tested. If 1990 weights understated Estonian growth over 1940–90, the Soviet-period series would arrive in 1937–40 systematically above the interwar series, which is weighted by the prices of 1938. It

[74] C. H. Feinstein and M. Thomas, *Making history count: a primer in quantitative methods for historians* (Cambridge: Cambridge University Press, 2002), 516.

[75] A. Gerschenkron, 'Problems in measuring long term growth in income and wealth', *Journal of the American Statistical Association* 52, no. 280 (1957): 453; Feinstein and Thomas, *Making history count*, 517.

[76] P. Jonas and H. Sardy, 'The Gerschenkron effect: a re-examination', *The Review of economics and statistics* 52, no. 1 (1970): 85.

[77] Joint Economic Committee, *Measures of Soviet*, 8; Joint Economic Committee, *USSR: Measures*, 23–4; Harrison, 'Soviet economic growth', 156.

does not: the four differences are small and of both signs. The structural change within industry, where the Gerschenkron effect originates, is moreover handled at a finer level than the aggregate weights suggest, since the 101 product series are aggregated within eleven branches with 1986 weights before the branches are combined. Third, the procedure is that of every comparable reconstruction of Soviet output: the CIA used 1970 weights for 1950–80 and 1982 weights for 1950–90, and Maddison found 'nothing untoward' in it; Kuboniwa's and Ponomarenko's reconstructions of Russian GDP for 1961–90 use, among their variants, the fixed structure of Russia's 1990 GDP.[78] Fourth, the result can be set beside series built on other weights. For 1950–68, the phase in which a late base would depress measured growth most, the new series gives 3.9 per cent a year for Estonia, the same as the 3.9 per cent of the East-Central European series that rest on Alton's multiple benchmarks and above the 3.4 per cent of the CIA's 1982-weighted series for the Union (Table 4); a series materially depressed by its weights would sit at the bottom of the bloc, and this one does not. A chain-linked alternative would require annual current-price value-added by sector, which exists for no Soviet republic, and a 1938-weighted alternative would require the sectors of 1990 to be valued at 1938 prices, when several of them did not exist. Nor has a benchmark at factor cost ever been constructed for a Soviet republic: the single-year study that Alton's project produced for Hungary runs to 254 pages, and a comparable exercise for a republic without price statistics of its own would be a monograph in itself.[79]

Fixed value-added weights

A related assumption is that the ratio of value-added to gross output within each sector remained at its benchmark-year level. The volume indices track gross output, inputs or employment and are combined with value-added weights, the method often loosely called single deflation.[80] Thomas and Feinstein's evaluation of this method found that it understated true value-added growth by between 0.15 and 1 percentage point a year in their simulations, depending on the pace of technical change, and they described it as 'the next best procedure' to double deflation where intermediate inputs cannot be observed.[81] Double deflation is not available for either

[78] Maddison, 'Measuring the Performance', 312; Ponomarenko, *Retrospective National Accounts*, 57, 62.
[79] T. P. Alton, *Hungarian national income and product in 1955* (New York: Columbia University Press, 1963).
[80] Thomas and Feinstein, 'The value-added', 443–4.
[81] Thomas and Feinstein, 427–8, 444.

period: the interwar statistical office recorded no intermediate consumption, and for the Soviet period, as Ponomarenko notes, the necessary price statistics 'practically did not exist'; the CIA itself double-deflated only agriculture and used gross output as the index of value-added everywhere else.[82]

The bias is bounded with the Swedish historical national accounts, which permit single- and double-deflated series to be compared over comparable phases of industrialisation.[83] In the Swedish case single-deflated series understated the GDP deflator by 4 per cent over the 75 pre-industrial years and by 17 per cent over the 75 years after 1875, and the Swedish evidence also shows that single deflation distorts short-run movements more than long-run ones, since the outputs of one sector are the inputs of another and the errors partly offset in the aggregate.[84] For the interwar period, when input costs outside agriculture moved little, the maximum error is about 2 per cent of the level of GDP. For the Soviet period, where the sustained shift towards intermediate-intensive heavy industry makes changing ratios likely, it is put at 14 per cent over the half-century (online appendix), with the sign of Thomas and Feinstein's simulations, an understatement of growth and hence an overstatement of the backcast early levels; 14 per cent over fifty years is about a quarter of a point a year, in the lower part of their range, as befits indices that track gross output product by product rather than by broad branch. The Soviet economy, moreover, adhered to its mass-production methods throughout its existence, so that input ratios are likely to have been more stable than in industrialising Sweden.[85] Two observations limit the practical concern. The error accumulates over the extrapolation and is largest in the earliest years of the Soviet-period series, precisely where the overlap shows the series within 4 per cent of an independent level. And the opposite sign cannot be excluded, since a rising degree of fabrication would raise value-added ratios if it outpaced the relative fall in intermediate-input prices, which is why the overlap, not the simulations, is decisive; Gerschenkron, confronting constant ratios over half a century in Jostock's German accounts, thought them 'not so very

[82]Ponomarenko, *Retrospective National Accounts*, 18–19; Joint Economic Committee, *USSR: Measures*, 45.

[83]L. Schön and O. Krantz, 'Swedish Historical National Accounts 1560-2010', *Lund Papers in Economic History*, no. 123 (2012): 19.

[84]H. Lobell et al., 'Swedish Historical National Accounts, 1800-2000: Principles and Implications of a New Generation', *The Scandinavian economic history review* 56, no. 2 (2008): 148–53.

[85]I. T. Berend, *Central and Eastern Europe, 1944-1993: detour from the periphery to the periphery* (Cambridge: Cambridge University Press, 1996), 195–7.

implausible' for just this reason.[86]

Other sources of bias

Table 7 lists the remaining sources of bias, the sectors they affect and the maximum impact of each on the level of GDP; the bounds for the Soviet period are derived in the same way as in the companion Lithuanian study, with the Estonian sectoral weights.

Table 7: Sources of potential bias and their maximum impact on the level of GDP

Period	Source of bias	Sectors affected	Max. impact (% of GDP)
Interwar, 1919–40	Service-sector value-added in the 1938 benchmark	Services (34% of GDP)	±3
	Fixed value-added weights ('single deflation')	All	±2
	Industrial output proxied by employment and real wages	Industry	±1
	Taxes less subsidies not tracked annually	All	±1
	No productivity adjustment in public services	Public services	±3
	Other services proxied by population	Other services	±1
	Military personnel known only for 1919–20 and 1939	Defence	±1
Soviet period, 1940–90	Service-sector allocation in the 1990 benchmark	Services	±4
	Taxes less subsidies absorbed in sectoral values	All	−0.5
	Fixed value-added weights ('single deflation')	All	−14
	Hidden inflation in comparable-price trade turnover	Trade	+2
	Size of Soviet forces stationed in Estonia unknown	Defence	±5
	No productivity adjustment in employment-based services	Services	−10
	Wartime movements 1943–5 partly borrowed from Lithuania	All (1941–5 only)	15–25

Sources: authors' assessment; see Section IV and the online appendix for the derivation of each bound.
Notes: signed entries indicate the direction in which the published series would be revised if the bias were corrected; ± entries are symmetric. The bounds are not additive.

Several deserve comment. The service sector of the 1938 benchmark, about one-third of GDP, rests on earnings and employment data and on Latvian structural ratios; an error of 10 per cent in the service estimate moves GDP by 3 per cent, and Norkus and Markevičiūtė's independent approximation, 5 per cent below the new benchmark, suggests that any error is of that order. In the 1990 benchmark the allocation of value-added among services depends on Bolotin's Union-wide parities and the 1987 input–output table; a reallocation of a quarter of service

[86] Gerschenkron, 'Problems in measuring', 453.

value-added between branches moves GDP by about 4 per cent. The Soviet-period trade index uses turnover at comparable prices, the one value series in the reconstruction; Kontorovich's estimates of hidden inflation in Soviet retail trade imply an upward bias of at most 2 per cent of GDP.[87] Defence after 1940 is unobserved, since the Soviet forces in Estonia were an all-Union institution of unknown size; their component of public services is interpolated between the 1940 and 1990 totals in step with civil employment, with a maximum bias of 5 per cent of GDP in either direction. The employment-based indices for health, education and administration make no allowance for productivity change, which understates service growth if productivity rose; the maximum effect is 10 per cent of the level. Finally, unregistered activity, perhaps 10–20 per cent of Soviet GDP in 1989, was of a size that OECD economies also displayed, so that its omission does not systematically impair international comparison.[88]

Taken together, the quantifiable biases imply a margin of about 6 per cent around the interwar levels and, around the Soviet-period levels, of 14 per cent for the largest single bias, or nearly a quarter in the earliest years if the two same-signed biases of Table 7, fixed value-added weights and unadjusted service productivity, were both at their maxima; the overlap shows that the realised error is far smaller. The war years are subject to a wider margin. The sub-period growth rates on which the argument of the following sections rests are robust to all of them: no plausible combination of the biases listed in Table 7 would raise growth in 1968–89 to the level of 1950–68, close the gap with Finland in 1938, or reverse Estonia's position relative to the Soviet average.

V | ESTONIAN GROWTH IN COMPARATIVE PERSPECTIVE

The series spans three regimes, the capitalist republic of 1919–40, the planned economy of 1940–91 and the market economy restored after 1991, and the literature on catch-up growth asks the same questions of each. Gerschenkron's late industrialiser borrows technology rather than inventing it and so grows faster than the leader, as Solow's model predicts from diminishing

[87] V. Kontorovich, 'Economists, Soviet Growth Slowdown and the Collapse', *Europe-Asia studies* (Abingdon) 53, no. 5 (2001): 688–9.

[88] B.-Y. Kim, 'Informal economy activities of Soviet households: size and dynamics', *Journal of comparative economics* 31, no. 3 (2003): 532–51; De Broeck and Koen, *The Great Contractions*, 5; F. Schneider, 'Dimensions of the Shadow Economy', *The independent review* 5, no. 1 (2000): 90–1.

returns to capital;[89] but Abramovitz and David showed that the potential is realised only where the follower has the 'social capability', the literacy, institutions and commercial organisation, to absorb what it borrows, and where the leader's technology is congruent with its endowments,[90] and North, and Acemoglu and Robinson, make convergence conditional on institutions that reward productive investment.[91] Estonia is an unusually clean test, because its social capability was high throughout by the standards of the European periphery, while its institutions were changed twice from outside. Tables 8, 9, 10 and 11 provide the comparative evidence.

Table 8: Estonian GDP per capita relative to comparators, selected years

	Estonia	Estonia as a percentage of					
Year	(Int$$_{1990}$$)	Finland	Sweden	Western core	USSR	Lithuania	Latvia
1913	2,058	97	72	51	146	126	87
1919	1,600	96	57	48	279	157	108
1929	2,113	78	52	45	152	134	90
1938	2,685	75	55	52	125	130	87
1940	2,771	86	57	52	129	159	—
1950	3,667	86	54	66	129	160	—
1968	7,348	91	64	70	141	136	—
1973	8,080	73	60	63	133	125	—
1980	8,785	68	59	61	137	127	—
1989	8,958	53	51	52	126	114	—
1995	6,277	39	35	33	145	121	125
2022	18,863	74	64	68	155	97	110

Sources: Estonia: see online appendix. Lithuania: A. Klimantas, 'Accounting for a Soviet republic: Lithuanian GDP time-series, 1940–1995' (Unpublished manuscript, University of Oxford, 2026); Latvia: A. Klimantas et al., 'Reinventing perished "Belgium of the East": new estimates of GDP for inter-war Latvia (1920–1939)', *Cliometrica* (Berlin) 18, no. 3 (2024): 765–835 (1940–89 not shown). Finland, Sweden, Western core and USSR: J. Bolt and J. L. van Zanden, 'Maddison-style estimates of the evolution of the world economy: a new 2023 update', Maddison Project Database, version 2023, *Journal of economic surveys* 39, no. 2 (2025): 631–671 (hereafter cited as MPD 2023) and online appendix. *Notes:* USSR within 1990 borders; 1995 and 2022 refer to the former Soviet Union aggregate.

[89] A. Gerschenkron, 'Economic backwardness in historical perspective', in *Economic backwardness in historical perspective: a book of essays*, ed. A. Gerschenkron (Cambridge: Belknap Press of Harvard University Press, 1962), 8; R. M. Solow, 'A contribution to the theory of economic growth', *The Quarterly journal of economics* 70, no. 1 (1956): 70–1.

[90] M. Abramovitz and P. A. David, 'Convergence and Deferred Catch-up: Productivity Leadership and the Waning of American Exceptionalism', in *The mosaic of economic growth*, ed. R. Landau et al. (Stanford: Stanford University Press, 1996), 21–62.

[91] D. C. North, *Institutions, institutional change and economic performance* (Cambridge: Cambridge University Press, 1990); Daron Acemoglu and James A. Robinson, *Why Nations Fail: The Origins of Power, Prosperity, and Poverty* (New York: Crown Publishers, 2012).

Interwar Estonia: recovery, depression and economic nationalism

The interwar record falls into two phases. The 1920s were a recovery: income per head grew by 2.8 per cent a year but regained its 1913 level only in 1929. The sectors that led, industry, construction and transport, were those most damaged, and the limits of the recovery were set by the permanent loss of the imperial market: pre-war industry had been built to serve Russia, and attempts to recover a Russian transit trade through Tallinn came to little.[92] Finland, less dependent on Russia and with forest industries that found Western buyers at once, grew by 5.1 per cent a year over the same decade (Table 4). Estonia's social capability was not in doubt, but the technology it had absorbed before 1914 was no longer congruent with the markets available to it.

The 1930s look different. The Depression was shallow, 8 per cent from peak to trough, and short; the recovery after 1933 was rapid, more than 4 per cent a year to 1938: Estonian income per head grew by 2.7 per cent a year between 1929 and 1938, when the Western core managed 1.1 per cent and East-Central Europe 0.1 per cent (Table 4). The timing is consistent with the exchange-rate explanation: Estonia clung to gold through the banking crisis of 1931–2 and recovered only after devaluing and pegging to sterling in June 1933, the pattern that Eichengreen and Sachs identified for Europe and that the Nordic economies displayed.[93] The content of the recovery was the policy that Kõll and Valge have called economic nationalism: state investment in oil shale, textiles and food processing, protection of the home market and the organisation of agricultural exports through co-operatives in imitation of Denmark.[94] The sectoral indices confirm that the acceleration was industrial and urban (Table 5).

Table 9 places the result in its regional setting. In 1938 Estonia's income per head, 2,685 Int$$_{1990}$$, was above that of Lithuania, Poland and Hungary, slightly below Czechoslovakia's and well below Latvia's, Finland's and the Nordic average. Its growth over 1929–38

[92] Kõll, 'The development gap', 193–210.

[93] Bernanke and James, 'The Gold Standard', 74; B. Eichengreen and J. Sachs, 'Exchange Rates and Economic Recovery in the 1930s', *The Journal of Economic History* 45, no. 4 (1985): 925–946; O. H. Grytten, 'Why was the Great Depression not so Great in the Nordic Countries? Economic Policy and Unemployment', *The Journal of European economic history* (Rome) 37, nos. 2-3 (2008): 369–403; V. Kuokštis, 'Searching for historical analogies in political economy: the Baltic states and the gold standard regime', *Journal of Baltic studies* 47, no. 3 (2016): 287–99.

[94] Kõll, 'Economic nationalism', 11–29; Norkus et al., 'Denmarkisation'; Klesment, 'Origins of economic nationalism', 118–35.

was exceeded in the region only by its Baltic neighbours, Finland and the Soviet Union (whose figures for the period are contested), and was far above that of the Western core and East-Central Europe. Table 11 shows why the result should be read as a success and why it was not a greater one. Estonia had the ingredients of social capability: adult literacy of around 92 per cent, higher than in any country of eastern or southern Europe except Latvia and Czechoslovakia and above Finland's, university enrolment per head among the highest in Europe, and land in the hands of smallholders after the reform of 1919. Yet 63 per cent of the labour force still worked in agriculture in 1934, a share as high as Finland's, though below Latvia's, and agriculture produced some 38 per cent of GDP in 1937 (Table 11). An economy of that structure, open to trade (Table 10) but exporting butter, bacon, timber and flax, could hardly grow faster than its agricultural productivity and the markets Britain and Germany allowed it; the industrial drive of 1934–9 was a response to that constraint, not an escape from it.

Table 9: GDP per capita in Estonia and European comparators, 1913–38

	Int$$_{1990}$$					Growth, % p.a.		
Country	1913	1919	1929	1932	1938	1919–29	1929–38	1913–38
Denmark	3,912	3,860	5,075	5,169	5,762	2.8	1.4	1.6
Sweden	2,874	2,825	4,063	3,968	4,901	3.7	2.1	2.2
Finland	2,111	1,658	2,717	2,550	3,589	5.1	3.1	2.1
Austria	3,465	2,259	3,699	2,940	3,559	5.1	−0.4	0.1
Latvia	2,361	1,481	2,351	2,286	3,099	4.7	3.1	1.1
Ireland	2,736	2,595	2,824	2,885	3,052	0.8	0.9	0.4
Czechoslovakia	2,096	1,588	3,042	2,680	2,803	6.7	−0.9	1.2
Estonia	2,058	1,600	2,113	2,105	2,685	2.8	2.7	1.1
Greece	1,455	1,274	2,342	2,289	2,677	6.3	1.5	2.5
Hungary	2,098	1,484	2,476	2,192	2,655	5.3	0.8	0.9
Poland	1,739	1,348	2,117	1,658	2,182	4.6	0.3	0.9
USSR	1,414	573	1,386	1,439	2,150	9.2	5.0	1.7
Lithuania	1,632	1,017	1,577	1,702	2,057	4.5	3.0	0.9

Sources: Estonia: see online appendix. Latvia and Lithuania: A. Klimantas et al., 'Reinventing perished "Belgium of the East": new estimates of GDP for inter-war Latvia (1920–1939)', *Cliometrica* (Berlin) 18, no. 3 (2024): 765–835; A. Klimantas, 'Lithuanian economy, 1919–1940: stagnant but resilient. The first inter-war GDP time-series estimates and their implications', *The Scandinavian economic history review* 72, no. 2 (2024): 129–158. Other countries: J. Bolt and J. L. van Zanden, 'Maddison-style estimates of the evolution of the world economy: a new 2023 update', Maddison Project Database, version 2023, *Journal of economic surveys* 39, no. 2 (2025): 631–671 (hereafter cited as MPD 2023). *Notes:* countries ordered by 1938 level.

Table 10: Trade openness (exports plus imports as a percentage of GDP), period averages

Country	1924–29	1930–34	1935–38
Estonia	33.4	25.6	31.5
Latvia	38.1	26.9	21.8
Lithuania	19.9	21.8	21.6
Finland	41.2	26.8	34.3
Denmark	51.8	35.2	36.3
Norway	56.1	35.6	41.1
Sweden	36.8	23.9	29.3
Poland	13.3	10.3	11.8
Czechoslovakia	36.8	23.6	26.5
Hungary	21.7	14.1	16.8
Germany	25.0	16.9	15.6
United Kingdom	36.9	23.3	25.6

Sources: see the online appendix for the compilation of trade and GDP series (Palgrave Macmillan Ltd, *International Historical Statistics*, 1st ed. 2013. (London: Palgrave Macmillan UK, 2013); J. Bolt and J. L. van Zanden, 'Maddison-style estimates of the evolution of the world economy: a new 2023 update', Maddison Project Database, version 2023, *Journal of economic surveys* 39, no. 2 (2025): 631–671 (hereafter cited as MPD 2023); national statistical yearbooks). *Notes:* unweighted averages of annual ratios; missing years omitted.

Table 11: Social capability and economic structure around 1930: Estonia, Finland and European comparators

Country	Literacy (% of adults) c. 1930	Agriculture (% of employment) c. 1930	University students (per 1,000) 1930	Agriculture (% of GDP) c. 1937
Sweden	99.6	36.0	1.6	13.0
United Kingdom	97.5	6.0	3.1	4.0
Germany	97.5	29.0	—	16.0
Denmark	97.5	35.3	2.7	18.0
Norway	97.5	35.5	2.2	13.0
Austria	97.5	39.0	3.2	14.0
Ireland	97.5	48.5	1.5	24.8
Czechoslovakia	96.0	37.5	—	24.0
Latvia	93.4	67.2	4.9	39.6
Estonia	92.0	63.0	3.5	37.6
Hungary	90.0	54.2	—	31.0
Finland	84.1	64.6	2.0	32.0
Italy	78.4	48.2	1.1	29.0
Lithuania	77.0	75.2	1.8	43.8
Poland	73.0	65.9	—	—
Bulgaria	69.0	71.8	—	52.1
Spain	68.0	51.9	1.7	22.0
Greece	58.1	56.6	1.1	54.4
Romania	57.1	77.2	—	—
Yugoslavia	55.0	79.7	—	41.0
USSR	50.0	73.0	—	29.0
Portugal	39.8	48.0	0.8	30.0

Sources: literacy: J. Cronin, *Literacy rates*, 5/7/13, 2013; D. Kirk, *Europe's population in the interwar years* (Geneva: League of Nations, 1946), 246–77; M. Krikščiūnas, 'Raštingumas', vol. XXIV, Lietuvių enciklopedija (Boston: Lietuvių enciklopedijos leidykla, 1961), 528. Agricultural employment: M. Morys, 'Economic growth and structural change in Central, East and South-East Europe, 1918–1939', ed. M. Morys, The economic history of Central, East and South-East Europe: 1800 to the present (Abingdon: Routledge, 2021), 181; Palgrave Macmillan Ltd, *International Historical Statistics*, 1st ed. 2013. (London: Palgrave Macmillan UK, 2013); W. W. Eason, 'Labor Force Materials for the Study of Unemployment in the Soviet Union', The Measurement and Behavior of Unemployment (Princeton: National Bureau Committee for Economic Research, 1957), 395, 409, 418. University students and agricultural shares of GDP: see online appendix, section A.7. *Notes:* countries ordered by literacy; a literacy value of 97.5 denotes near-universal literacy where the source gives no census figure. '—': not available.

Under Soviet rule: collectivisation, industrialisation and stagnation

The Soviet period opens with a decade of destruction. Income per head fell by about a third between 1940 and 1944 on the present estimates, and the resident population by a fifth between 1939 and 1945, of which about six points reflect the transfer of the districts east of Narva and of Petseri to Russia and the rest war deaths, deportation and flight.[95] The recovery of industry and construction after 1945 was rapid, as Soviet reconstruction everywhere was: both regained their 1938 level by 1948, and industry grew by around 20 and construction by around 14 per cent a year between 1945 and 1953 (Tables 5 and A2). The war had destroyed people rather than plant: with a fifth of the population gone, productive capacity stood underused, and the first years of recovery, in which the investment ratio was still below the Soviet mean even on official data, rested on putting labour, increasingly immigrant labour directed from Russia, back to work on existing capital, the mechanism that Vonyó and Markevich identify across post-war East-Central Europe.[96] Agriculture tells a different story. Livestock output, halved by the war, was still 38 per cent below its 1938 level in 1950, the year after collectivisation; crop output recovered sooner, but the agricultural sector as a whole stood at 60–65 per cent of its 1938 level in 1945–6, fluctuated between 77 and 93 per cent of it in 1947–57, and exceeded it only in 1959. The index shows a shallower trough than Mertelsmann's archival estimate, which puts output after collectivisation at perhaps 40 per cent of the 1939 level, but the two agree that the sector did not recover for a decade, and that the collective and state farms produced little more than a third of output in the early 1950s, the rest coming from private plots, which still supplied 28 per cent in 1960.[97] Nothing comparable to the Soviet famine of 1932–3 occurred, probably because the Baltic collectivisation of 1949 did not coincide with a procurement crisis and because, as Harrison has argued for Soviet agriculture generally, the system had by then learned to extract

[95] Taagepera, 'Soviet Collectivization Of Estonian Agriculture: The Taxation Phase', 265; Õispuu, *Name list*, 12.

[96] R. J. Misiunas and R. Taagepera, *The Baltic states: years of dependence, 1940-1990*, Expanded and updated ed. (London: Hurst, 1993), 108; Vonyó and Markevich, 'Economic growth and structural', 290; Kasekamp, *A history*, 126–8.

[97] O. Mertelsmann, *Stalinist Restructuring in Estonia: From Market to Command Economy* (Lausanne: Peter Lang, 2024), 156–8; Feest, 'Collectivization', 79–109; Klesment, 'Estonian economy under Soviet rule', 257; K. G. Hansen, 'Estonian agriculture: from a Soviet past to an uncertain future', *Oxford agrarian studies* 22, no. 2 (1994): 160.

the surplus without destroying the herds.[98] The cost of war and collectivisation together was a lost decade and a permanent shift of effort towards animal husbandry, in which Estonia came to lead the Union in output per head, while crop output never again exceeded the level it reached in 1968.[99]

Between 1950 and 1968 Estonia experienced what Gerschenkron would have recognised as a classic burst of state-led industrialisation. Total GDP grew by 5.4 per cent a year and income per head by 3.9 per cent, faster than in Finland, the Nordic countries or the Western core, though not than the Western periphery, whose Iberian and Greek members were in the middle of their own industrialisation (Table 4). The engine was investment directed from Moscow into three complexes. The first was the oil-shale basin of the north-east, which fed two of the largest thermal power stations in the Union and raised electricity output from 2 billion kilowatt-hours in 1960 to 12 billion in 1970, two-thirds of it consumed outside the republic. The second was cotton textiles, expanded in Estonia rather than in cotton-growing Uzbekistan because Estonia had the skilled labour and the mills; the third, machine-building and electronics for the Union market.[100] The pace was already slackening: industrial output had grown by 15 per cent a year in 1948–55 and grew by 9 per cent a year in the decade that followed, and the sovnarkhoz reform of 1957–65, which briefly placed industrial management in Tallinn, was Khrushchev's response to that deceleration across the Union rather than its cause. Baltic historiography has judged the reform favourably, perhaps more for the autonomy it conferred than for its results; it saw official industrial output double in seven years (the physical index used here rises by 80 per cent over 1957–64), with 70 per cent of investment going to industry and the share of light industry in the total deliberately reduced, and it was superseded in 1965 by the Kosygin reform and the recentralisation of enterprises.[101] The labour came partly from a countryside that collectivisation had released and partly from immigration: Estonia's population grew by half between 1950 and 1989 while the Estonian share of it, close to 97 per cent at the end of the war, had fallen to

[98] M. Harrison, 'Soviet agriculture and industrialisation', in *Agriculture and economic growth from the eighteenth century to the present*, ed. J. A. Davis and P. Mathias (Oxford: Blackwell, 1996), 192–208.

[99] Misiunas and Taagepera, *The Baltic states*, 234.

[100] Misiunas and Taagepera, 239; Ahde and Rajasalu, *Economic Structure of Estonia*, 41–7; M. Pihlamägi, 'Administrative shakeup in industry and construction: the 1957–1965 'Sovnarkhoz' reform in the Estonian SSR', *Nordost-Archiv. Zeitschrift für Regionalgeschichte* 22 (2013): 248–58.

[101] Pihlamägi, 248–52; A. Purre, ''New Deal' in Soviet industrial administration', *Lituanus* 11, no. 4 (1965): 67–70; Misiunas and Taagepera, *The Baltic states*, 189.

three-quarters by the census of 1959 and to little more than 60 per cent by that of 1989.[102]

How did this growth compare with that of the Union into which Estonia had been absorbed? Table 8 gives the answer, and it is not the one that either Soviet or nationalist accounts have assumed. Estonia entered the Union in 1940 with income per head 129 per cent of the Soviet average; it stood at 141 per cent in 1968 and at 126 per cent in 1989. The gap neither closed nor widened appreciably over half a century: Estonia grew with the Union, somewhat faster in the 1950s and 1960s and somewhat slower thereafter, and remained one of its two or three richest republics throughout. The result matches Soviet-era estimates that placed Estonia at 137–145 per cent of the Union average in 1960–70 by net material product, and agrees in direction, if not in magnitude, with the CIA's finding that Estonian national income per head grew faster than the Union's in 1961–9, 6.3 against 5.5 per cent a year on NMP-based data that overstate growth.[103] Did Moscow buy the loyalty of its western borderlands with investment? Investment per head in Estonia was above the Union average in the 1960s, second only to Kazakhstan, but the republic's share of investment in its own national income was below the Union average. Van Selm and Dölle's reconstruction of inter-republican transfers finds Estonia's net balance close to zero from 1966 to 1978, becoming a net recipient of 4–6 per cent of national income only after 1979 and of 13 per cent in 1988.[104] For the years after 1966 for which transfers can be measured, Estonia's high income was not the product of net transfers; and over the period as a whole it was inherited rather than created, since the republic was richer than the Union at annexation and remained so in roughly the same proportion.

The deceleration after 1968 was general and, by the late 1970s, complete. Income per head grew by 0.9 per cent a year over 1968–89 and by 0.6 per cent after 1973, and the sectoral peaks cluster in a single decade: food processing in 1976, construction in 1977, electricity and oil-shale output in 1979–80, heavy industry in 1980.[105] The turning point coincided with the Prague

[102] Parming and Järvesoo, *A Case study*, 158–217.

[103] Schroeder, 'Regional differences', 169, 174; I. S. Koropeckyj, 'National income of the Baltic Republics in 1970', *Journal of Baltic studies* (Brooklyn) 7, no. 1 (1976): 64–71; Central Intelligence Agency, *Soviet regional development in 1960–69: trends and implications*, USSR ECON 72-04 (Office of Economic Research, 1972), 8–10; Bohnet and Penkaitis, 'A comparison', 35–8.

[104] Central Intelligence Agency, *Soviet regional development*, 22–9; I. Adirim, 'Realities of economic growth and distribution in the Baltic states', *Journal of Baltic studies* 19, no. 1 (1988): 53; G. van Selm and E. Dölle, 'Soviet interrepublican capital transfers and the republics' level of development, 1966–1991', *MOCT-MOST*, no. 1 (1993): 136–40.

[105] Electricity peaked in 1979 and fell by 9 per cent to 1985; oil-shale mining levelled off after the arrival of

Spring and the end of Moscow's tolerance of regional economic autonomy, and with the year in which the Union as a whole ceased to converge on Western levels, ahead of most other European communist economies.[106] The sources point to three causes. The Union-wide investment slowdown imposed by the tenth and eleventh five-year plans fell on a republic whose growth had been investment-led and whose labour reserves were exhausted: further expansion required immigration or capital deepening, and the planners, whose investment effort shifted eastwards to Siberia in the 1970s, increasingly withheld both; on official data the investment ratio in Estonia fell below the Soviet average by the mid-1970s and stayed there.[107] The oil-shale complex, built on a resource that was expensive to mine and polluting to burn, lost its rationale once Siberian gas reached the Baltic.[108] And the gains in reported output after 1975 were increasingly nominal: Finnish and Estonian economists who compared the two economies in 1992 concluded that the rapid growth of Estonian industry to 1975 reflected real investment, but that 'the further growth of production can be explained mainly by the change in prices'.[109] The physical indices used here confirm that judgement: industrial output grew by 1.4 per cent a year after 1968, and the branches that continued to grow, textiles and other light industry, did so slowly, while cotton cloth output was flat from 1970.[110] The textile index itself fell by 14 per cent between 1972 and 1975 and did not regain its 1972 level until 1985 (Table A2), the one sharp break in a branch that otherwise grew throughout the Soviet period; the sources consulted offer no explanation, and the authors record it rather than interpret it. Three further irregularities in the indices deserve mention. Construction, which had grown by about 9 per cent a year since 1950, peaked in 1977 and stagnated for the rest of the Soviet period as the Union-wide investment squeeze took hold; crop output fell by a quarter in each of the poor harvests of 1978 and 1981 that affected the Union as a whole; and the housing index, which is derived from construction activity and

natural gas (Misiunas and Taagepera, *The Baltic states*, 239; R. Taagepera, *Estonia: return to independence*, First published 1993 (London: Routledge, 2018)).

[106] Misiunas and Taagepera, *The Baltic states*, 203, 227–8.

[107] P. Hanson, *The rise and fall of the Soviet economy: an economic history of the USSR from 1945* (Harlow: Pearson Education, 2003), 128–63; E.-J. Kolde, 'Structural integration of Baltic economies into the Soviet system', *Journal of Baltic studies* 9, no. 2 (1978): 164–76; L. Dienes, 'Investment priorities in Soviet regions', *Annals of the Association of American Geographers* 62, no. 3 (1972): 441; CIA, *Economic Development in the Soviet Republics, 1970-89* (Washington, D.C.: CIA Office of Soviet Analysis, July 1990), 2.

[108] Mertelsmann, *Stalinist Restructuring*, 188–9.

[109] Ahde and Rajasalu, *Economic Structure of Estonia*, 39–40.

[110] Ahde and Rajasalu, 41.

population and had grown by between 2 and 3 per cent a year since the 1950s, stopped growing in 1986, when completions began to fall from 13,500 dwellings a year in 1985 to 5,200 in 1991; its slight decline thereafter reflects the construction of the proxy rather than a physical loss of stock.[111]

Read through the lens of growth theory, the Soviet episode is coherent. The catch-up of 1950–68 is what the models predict when a capital-poor economy with high social capability is made to invest, and mid-century mass-production methods were congruent with Estonia's literate and urbanised labour force; the stagnation that followed is equally what they predict when the institutional condition fails. Extensive growth exhausted its inputs, and the system could not replace them with the productivity growth that flexible production, foreign technology and competition were delivering across the Gulf: the share of services in Estonian GDP was 42 per cent in 1990, barely above its interwar level, against about 60 per cent in Finland.[112] Estonia's income relative to the Western core, which had risen from 66 per cent in 1950 to 70 per cent in 1968, fell back to 52 per cent by 1989, almost exactly its level of 1913. Two decades of Soviet growth had been cancelled by two decades of Soviet stagnation; by 1989 Greece, the poorest capitalist economy in Western Europe, had overtaken Estonia, one of the richest parts of the socialist world after Slovenia (Table 8).

Transition and recovery

The transition compressed into four years a fall in income per head of 34 per cent, deeper than the Depression and comparable only to the two world wars. The sectoral indices to 1991 show the collapse beginning in Soviet-era industry and construction, which were already shrinking at −9.1 and −9.9 per cent a year before the rouble was abandoned. After 1991 the modern series records the loss of eastern markets and cheap Soviet inputs, the currency board of 1992 and the stabilisation that followed.[113] By comparison with the other Soviet republics the fall was

[111] Hanson, *Rise and fall*, 134–50; Ahde and Rajasalu, *Economic Structure of Estonia*, 60, 66; World Bank, *Estonia*, 216.

[112] Table 2; Ahde and Rajasalu, *Economic Structure of Estonia*, 11–12; Klimantas, 'Accounting for a Soviet republic'.

[113] World Bank, *Estonia*, 1–8; Norkus, *Post-Communist Transformations*, 164–6; I. Shteinbuka, 'The Baltics', in *Economic consequences of Soviet disintegration*, ed. J. Williamson (Washington, DC: Institute for International Economics, 1993), 482–509.

contained, probably because Estonia had less heavy industry to lose, 41 per cent of industrial value-added in 1990 against 48 in Latvia and 55 in Lithuania, and reoriented trade towards Finland and Sweden faster than any other successor state.[114] Recovery from 1995 at 4.2 per cent a year carried Estonia from 33 per cent of the Western core in 1995 to 68 per cent in 2022, close to the 70 per cent reached in 1968 at the height of Soviet growth. Whether the market-era catch-up will continue or stall in the middle-income trap that Staehr describes is a question for another paper; the long series shows that Estonia converged on Western Europe under capitalism twice, in the 1930s and after 1995, and under planning once, briefly, before diverging again.[115]

VI | A FINNISH MIRACLE THAT NEVER WAS

The comparison with Finland has shaped the way Estonians and their historians think about the twentieth century. The two peoples speak related languages, were shaped by Lutheran schooling and Swedish law, emancipated their peasantries early and built co-operative agrarian economies after 1918; in the Soviet period Finland, visible on television from Tallinn, became 'proof of what Estonia would have been if the country had not ended up under the Soviet occupation'.[116] The scholarly version of the argument has two steps. The first is that the two countries stood at the same level before 1940: Dellenbrant wrote that their levels of development were 'close to equal around 1940', Soviet-era comparisons of real wages put Estonian workers ahead of Finnish ones as late as 1966, and the White Book's estimate of the losses of occupation rests on the assumption that Estonia would have followed the Finnish path.[117] The second is that the gap observed in 1991, when Estonian living standards were put at 40 per cent of Finland's, measures the cost of Soviet rule.[118] Sceptics have questioned both steps. Kõll argued in 1994 that the belief in Baltic–Nordic parity before the war was 'misleading'; an Estonian economist

[114]Bolotin, 'Former Soviet Union', 184–6; T. Grennes, 'The economic transition in the Baltic countries', *Journal of Baltic studies* (Brooklyn) 28, no. 1 (1997): 22; D. J. Smith, 'Estonia: Independence and European Integration', ed. D. J. Smith et al., The Baltic states: Estonia, Latvia and Lithuania (London: Routledge, 2002), 117; Korhonen, 'Progress', 443–4.

[115]K. Staehr, 'Economic growth and convergence in the Baltic states: caught in a middle-income trap?', *Inter-economics* 50, no. 5 (2015): 274–80.

[116]Lepp and Pantti, 'Window to the West', 83.

[117]Dellenbrant, 'Estonia's economic development', 206; Klesment, 'Estonian economy under Soviet rule', 251, 257; Kukk, 'Economic damage', 141–74.

[118]Dellenbrant, 'Estonia's economic development', 206.

of the 1930s had already written that 'Denmark, Sweden, Finland, and even Latvia are ahead of us regarding their wealth'; and Norkus, working from the benchmark estimates, concluded that the gap 'cannot only be considered as the legacy of the Soviet occupation'.[119] The new series allows the two steps to be examined year by year.

Table 8 gives Estonian income per head as a percentage of Finland's at the benchmark years, and Table 4 the growth rates between them. In 1913 the two countries were indeed close: Estonia stood at 97 per cent of the Finnish level. In 1919, after wars that had damaged both, Finland the more severely in 1918, it stood at 96 per cent. The parity did not survive the 1920s. Finland grew by 5.1 per cent a year to 1929 and Estonia by 2.8 per cent, and by 1929 Estonia was at 78 per cent of the Finnish level; the 1930s, in which Estonia's growth of 2.7 per cent a year was close to Finland's 3.1 per cent, left the ratio at 75 per cent in 1938. Clark's 90 per cent, which Valge adopted and which has anchored the literature since, is thus too high by a sixth; Bocharnikova's finding that the purchasing power of wages in Estonia was only 4 per cent below Finland's in 1938 refers to urban wages alone and says nothing about the nearly two-thirds of the Estonian labour force on the land.[120] The difference in 1938 was not one of social capability. Table 11 shows Estonian literacy above Finnish, Estonian university enrolment per head well above Finnish, and the same share of the labour force in agriculture; Table 10 shows both economies open, though Finland more so. The difference is better explained by resources and markets. Finland's forests, sawmills and pulp mills gave it an export staple for which Western demand recovered quickly after 1920 and which carried a growing engineering sector with it; Estonia's butter and bacon faced Danish competition and British quotas, and its industry had lost the only large market it had ever served.[121]

The second step of the argument fares no better, though for a different reason. Soviet rule did open a gap, but the timing is not what the argument supposes. Finland lost the Second World War and paid reparations; Estonia was fought over twice and collectivised. In 1950 the ratio stood at 86 per cent, higher than in 1938, and in 1968, after the two decades of Soviet industrialisation

[119]Kõll, 'The development gap', 201; Klesment, 'Origins of economic nationalism', 131; Norkus, 'A comparison', 18; Norkus, *Post-Communist Transformations*, 154.

[120]Bocharnikova, 'Economic well-being', 81–2.

[121]R. Hjerppe and J. Jalava, 'Economic Growth and Structural Change: A Century and a Half of Catching-up', ed. J. Ojala et al., The Road to Prosperity: An Economic History of Finland (Helsinki: Suomalaisen Kirjallisuuden Seura, 2006), 184, 198–200; Kõll, 'The development gap', 193–210.

examined above, at 91 per cent, the highest level since 1919. Measured against Finland, the first half of Soviet rule was a period of convergence. The divergence that the literature attributes to occupation is a phenomenon of its second half: between 1968 and 1989 Finland grew by 3.6 per cent a year, half as fast again as the Western core, and Estonia by 0.9 per cent, so that the ratio fell to 73 per cent in 1973 and 53 per cent in 1989. Those who lived through it dated the divergence correctly: Dellenbrant, drawing on Kukk's calculations, placed its beginning in the late 1960s.[122] What the new series adds is the magnitude. Expressed in logarithms, the Estonian shortfall against Finland was 0.29 in 1938 and 0.63 in 1989; almost half of the gap at the end of Soviet rule had opened before it began, and the whole of the increase after 1938 occurred after 1968.

Two conclusions follow. The first concerns the use of Finland as a yardstick. Finland is a revealing comparator for Estonia, because the two economies had similar endowments of human capital, though not of natural resources or markets, so that part of their divergence can be read as institutional. But it is not a measure of what Estonia lost. The two economies were already a quarter apart in 1938, and Finland's growth after 1968, 3.6 per cent a year, was exceptional by the standards of the Western core (2.4 per cent) and of the Nordic countries (2.5 per cent), driven by a forest-products and engineering complex that Estonia never possessed and by access to a Soviet market on favourable bilateral terms that Estonia, inside the Union, could not exploit.[123] The Western periphery as a whole, the group of poorer Western economies to which Estonia belonged by income in 1938, grew by 3.0 per cent a year over the same years (Table 4); its Iberian members, starting from levels below Estonia's of 1950, grew as fast as Finland, Greece and Ireland somewhat more slowly. Measured against the core or the periphery rather than against Finland, Estonia's shortfall in 1968–89 is between one and a half and two points a year rather than more than two and a half: still large, but a different magnitude from the one the White Book implies, and one that cannot be read off the Finnish series alone. The second concerns the Soviet record itself. The years in which Estonia kept pace with Finland were the years of forced industrialisation, high investment and immigration; the years in which it fell behind were those in which the planned economy, having exhausted the sources of extensive growth, could neither

[122] Dellenbrant, 'Estonia's economic development', 205.
[123] Hjerppe and Jalava, 'Economic Growth', 184–200; Norkus, 'A comparison', 23–4.

raise productivity nor change its structure. The gap with Finland in 1991 is therefore likely to overstate the cost of occupation. The part of it that can be charged to the system depends on the yardstick: measured against the core, the cost is confined to 1968–89; measured against the periphery, it was already 1.3 points a year in 1950–68 and widened to 2.1 thereafter. On either yardstick it is concentrated in, though not confined to, the last two decades, and on either it is smaller than the Finnish comparison implies: a less comforting conclusion than the one the White Book drew, and a more exact one.

VII | CONCLUSION

This paper has constructed the first annual, internationally comparable series of Estonian GDP per capita for 1919–95 and joined it to existing estimates to give a continuous series for 1913 and 1919–2022 in 1990 and 2011 international dollars. It rests on a new 1938 benchmark from a direct price comparison with Sweden, a recalculated production-side series for 1919–40, and a new series for the ESSR for 1937–95 built from physical output, transport and employment data and anchored to Bolotin's 1990 benchmark. Built from different sources, with different weights and from benchmarks half a century apart, the two new series agree within 4 per cent in every year of their 1937–40 overlap, and the Soviet-period series meets a modern series derived from the 2011 ICP round within 1 per cent at the 1991 splice. The series supersedes the three partial estimates that preceded it, lowering late-Soviet income by about a quarter relative to the Maddison Project's NMP-based figures and raising it by a similar margin relative to the only earlier long series, and it is ready for the Maddison Project Database.

Three findings stand out. First, Estonia's interwar growth was respectable but not exceptional: 2.8 per cent a year in the 1920s, as the country recovered from war and the loss of its imperial market, and 2.7 per cent in the 1930s, when devaluation and a state-led industrial drive produced a rapid recovery from the Depression. Second, Soviet rule brought two decades of rapid growth and two of stagnation: 3.9 per cent a year in 1950–68, faster than in Finland or the Western core, as investment directed from Moscow built an oil-shale, textile and engineering economy on an unusually literate labour force, and 0.9 per cent in 1968–89, as the sources of extensive growth ran out. Estonia's income relative to the Western core rose from two-thirds in

1950 to seven-tenths in 1968 and fell back to half by 1989, while within the Union it remained what it had been at annexation, a republic between a fifth and a half richer than the average. Third, the transition cut income per head by a third in four years, and recovery after 1995 carried Estonia back, by 2022, close to the share of Western European income it had briefly reached in 1968.

The comparison with Finland, which gives the paper its title, needs restating. The two countries were close in 1913 and 1919, but by 1938 Estonia stood at three-quarters of the Finnish level, and almost half of the gap of 1989 had opened under independence, not occupation; the further divergence was concentrated after 1968, Estonia having kept pace with Finland in the preceding two decades. Finland shows what institutions can do with a given endowment of human capital, but not what Estonia would have become, because the Finnish path depended on resources and markets that Estonia did not have. The costs of Soviet rule were real, and the series locates most of them in the stagnation of 1968–89; measured against the Western core or periphery rather than against a Finland whose growth in those decades was exceptional, they are smaller than the Finnish comparison implies, and they are not the costs of a Finnish miracle that Estonia was denied.

Several extensions are open: a dedicated study of the 1938 service sector of the kind now available for Lithuania and Latvia; a Soviet-period benchmark built from the Estonian archives with republican rather than Union-wide prices, which would test the level to which the volume indices are anchored; expenditure-side cross-checks for a few interwar years; and the application of the procedure to the other non-Russian republics, whose national accounts remain the largest gap in the quantitative economic history of the Soviet Union.

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STATISTICAL APPENDIX

This appendix reports the annual series on which the paper rests. Table A1 gives GDP per capita in 1990 and 2011 international dollars and the mid-year population for 1913 and every year from 1919 to 2022; Tables A2 and A3 give the constant-price volume indices of sectoral value-added for 1919–95 from which the GDP series of 1919–91 is built, with 1938 equal to 100. The indices for 1992–5 are those of the Soviet-period series and are superseded, in the final series, by the modern series from 1991 (Section IV). The online appendix, available as an Excel workbook, contains the same series together with the 1938 and 1990 benchmarks by sector, the overlapping estimates at the two splicing points, the prices and weights of the 1938 comparison with Sweden, the product-by-product composition of the industrial index, and a source list for every series by year, keyed to the sections A.1–A.7 referred to in Tables 3 and 11. The series to 1944 refer to the borders of the republic and the series from 1945 to present-day borders (Section II).

Table A1: Estonian GDP per capita and population, 1913 and 1919–2022

Year	Int$$_{1990}$$	Int$$_{2011}$$	Population	Year	Int$$_{1990}$$	Int$$_{2011}$$	Population
1913	2,058	3,280	1,164.0	1971	8,002	12,755	1,377.0
1919	1,600	2,550	1,064.0	1972	7,896	12,587	1,392.5
1920	1,475	2,351	1,067.8	1973	8,080	12,879	1,406.0
1921	1,673	2,666	1,087.1	1974	8,217	13,098	1,418.2
1922	1,756	2,800	1,102.4	1975	8,419	13,419	1,429.4
1923	1,913	3,050	1,110.8	1976	8,623	13,745	1,439.6
1924	1,895	3,020	1,115.6	1977	8,601	13,709	1,450.2
1925	2,038	3,249	1,117.0	1978	8,474	13,508	1,460.2
1926	2,030	3,235	1,116.8	1979	8,523	13,585	1,468.3
1927	2,043	3,256	1,115.6	1980	8,785	14,003	1,477.2
1928	2,046	3,262	1,115.7	1981	8,585	13,684	1,487.7
1929	2,113	3,368	1,115.7	1982	8,554	13,635	1,498.4
1930	2,282	3,638	1,116.1	1983	8,775	13,988	1,508.7
1931	2,176	3,469	1,118.4	1984	8,932	14,238	1,518.6
1932	2,105	3,356	1,121.5	1985	8,728	13,913	1,528.8

continued

Table A1 (continued)

Year	Int$$_{1990}$$	Int$$_{2011}$$	Population	Year	Int$$_{1990}$$	Int$$_{2011}$$	Population
1933	2,186	3,485	1,124.3	1986	8,919	14,216	1,540.2
1934	2,353	3,751	1,126.4	1987	8,833	14,079	1,552.2
1935	2,355	3,754	1,128.9	1988	8,879	14,153	1,561.9
1936	2,492	3,972	1,130.0	1989	8,958	14,280	1,568.1
1937	2,722	4,338	1,130.7	1990	8,425	13,429	1,569.2
1938	2,685	4,280	1,132.5	1991	7,532	12,006	1,561.3
1939	2,736	4,361	1,140.4	1992	6,069	9,675	1,533.1
1940	2,771	4,418	1,122.7	1993	5,870	9,357	1,494.1
1941	2,390	3,809	1,049.1	1994	5,899	9,402	1,462.5
1942	2,382	3,797	1,007.0	1995	6,277	10,005	1,436.6
1943	2,328	3,710	981.9	1996	6,685	10,655	1,415.6
1944	1,909	3,044	917.8	1997	7,644	12,184	1,399.5
1945	2,656	4,234	903.1	1998	8,053	12,836	1,386.2
1946	2,833	4,516	938.7	1999	7,995	12,744	1,390.2
1947	3,123	4,978	990.2	2000	8,759	13,961	1,397.0
1948	3,323	5,297	1,026.3	2001	9,344	14,894	1,388.1
1949	3,474	5,538	1,026.1	2002	10,040	16,004	1,379.3
1950	3,667	5,846	1,036.4	2003	10,871	17,328	1,370.7
1951	3,709	5,912	1,061.6	2004	11,680	18,618	1,362.6
1952	3,758	5,990	1,083.1	2005	12,867	20,509	1,354.8
1953	3,747	5,973	1,106.5	2006	14,207	22,645	1,346.8
1954	3,787	6,036	1,128.9	2007	15,353	24,473	1,340.7
1955	3,951	6,298	1,144.2	2008	14,604	23,279	1,337.1
1956	4,243	6,763	1,157.9	2009	12,492	19,912	1,334.5
1957	4,460	7,109	1,171.9	2010	12,826	20,445	1,331.5
1958	4,727	7,535	1,187.8	2011	13,800	21,997	1,327.4
1959	5,010	7,986	1,201.6	2012	14,294	22,784	1,322.7
1960	5,319	8,478	1,211.5	2013	14,557	23,204	1,318.0
1961	5,513	8,788	1,225.1	2014	15,046	23,983	1,314.7
1962	5,547	8,841	1,241.6	2015	15,354	24,474	1,314.0
1963	5,840	9,309	1,258.9	2016	15,807	25,196	1,314.4
1964	6,150	9,803	1,277.1	2017	16,726	26,662	1,316.1

continued

Table A1 (continued)

Year	Int$$_{1990}$$	Int$$_{2011}$$	Population	Year	Int$$_{1990}$$	Int$$_{2011}$$	Population
1965	6,466	10,308	1,294.6	2018	17,313	27,596	1,320.6
1966	6,613	10,541	1,308.6	2019	17,883	28,506	1,325.1
1967	7,013	11,178	1,318.9	2020	17,730	28,262	1,329.5
1968	7,348	11,712	1,331.2	2021	19,134	30,499	1,330.9
1969	7,442	11,863	1,345.2	2022	18,863	30,067	1,348.8
1970	7,688	12,255	1,360.1				

Sources: see online appendix. *Notes:* GDP per capita in international Geary–Khamis dollars of 1990 and of 2011 (Int$$_{2011}$$ = Int$$_{1990}$$ × 1.594, Bolt and Zanden, 'The long view', 12); population in thousands, mid-year, within the borders of the republic to 1944 and present-day borders thereafter. No estimates for 1914–18.

Table A2: Sectoral volume indices, Estonia 1919–95: primary and secondary sectors

Year	Crops	Livestock	Forestry, fishing	Primary	Food	Textiles	Other light	Heavy	Industry	Construction	Secondary
1919	60.6	41.5	60.0	50.5	—	—	—	—	20.4	17.1	20.1
1920	62.7	58.6	72.5	61.2	—	—	—	—	26.3	17.2	25.4
1921	77.6	66.9	86.0	72.6	—	—	—	—	35.3	17.5	33.6
1922	74.6	72.0	87.7	74.1	—	—	—	—	42.7	17.8	40.3
1923	72.2	74.9	93.4	75.1	—	—	—	—	55.1	61.7	55.7
1924	71.0	78.4	108.1	77.4	—	—	—	—	54.8	18.4	51.3
1925	83.3	87.6	116.1	87.9	—	—	—	—	53.8	36.2	52.1
1926	71.8	91.3	95.9	83.7	—	—	—	—	57.7	39.6	56.0
1927	76.5	93.0	106.0	87.2	—	—	—	—	54.1	15.7	50.5
1928	64.3	93.7	96.2	82.0	—	—	—	—	55.6	51.9	55.2
1929	78.2	87.4	94.6	84.2	—	—	—	—	60.7	51.8	59.8
1930	100.1	92.9	85.5	95.3	—	—	—	—	63.4	46.1	61.7
1931	79.3	96.9	80.3	88.6	—	—	—	—	62.1	39.1	59.9
1932	79.8	96.2	111.8	90.6	—	—	—	—	55.0	37.2	53.3
1933	92.5	94.9	111.8	95.1	—	—	—	—	61.1	36.8	58.8
1934	104.8	93.2	100.1	98.4	—	—	—	—	75.2	38.7	71.7
1935	90.2	93.9	98.7	92.7	—	—	—	—	78.8	55.7	76.6
1936	87.2	89.7	100.0	89.4	—	—	—	—	98.5	72.1	96.0
1937	101.7	100.2	98.6	100.7	—	—	—	—	104.4	102.6	104.2
1938	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
1939	101.5	99.7	96.7	100.2	104.4	110.4	96.7	89.1	107.1	111.7	107.6
1940	104.4	85.3	83.4	92.9	101.2	113.1	85.1	81.4	108.3	102.5	107.7
1941	76.9	62.6	82.4	68.8	78.9	64.0	75.9	67.3	83.8	88.9	86.3
1942	77.0	48.8	81.5	61.6	61.7	22.5	66.7	60.7	64.4	80.3	69.9

continued

Table A2 (continued)

Year	Crops	Livestock	Forestry, fishing	Primary	Food	Textiles	Other light	Heavy	Industry	Construction	Secondary
1943	64.2	47.1	80.5	55.1	55.7	30.1	65.4	57.9	63.4	69.8	66.1
1944	57.4	39.1	78.4	47.8	42.1	20.1	65.3	34.5	15.5	19.0	16.7
1945	78.9	44.0	78.6	59.9	45.5	32.7	68.9	51.6	61.2	87.3	69.7
1946	86.4	47.1	90.9	65.1	67.4	47.8	84.8	58.7	78.5	95.9	84.6
1947	118.4	49.9	104.2	81.0	78.4	73.5	99.1	80.8	99.1	117.6	105.8
1948	131.5	55.9	102.2	89.9	104.6	96.9	112.5	103.9	122.9	136.2	128.2
1949	117.2	59.2	107.4	85.5	117.1	132.4	127.8	141.1	150.9	162.4	156.0
1950	113.6	62.0	111.5	85.5	130.3	172.0	138.7	182.7	179.4	187.0	183.7
1951	106.9	62.3	114.9	82.7	150.8	232.3	140.4	223.6	210.6	209.2	212.5
1952	105.3	66.3	107.5	83.9	163.7	269.8	136.7	266.1	232.9	231.4	235.0
1953	97.3	64.5	110.8	79.5	179.9	304.2	155.0	297.3	261.0	252.5	261.3
1954	88.7	69.0	131.7	78.6	203.7	326.3	170.5	322.2	285.2	269.3	283.6
1955	86.0	67.5	165.5	77.4	219.0	379.2	207.0	380.0	332.5	320.9	332.6
1956	109.8	70.5	177.9	89.8	229.7	382.3	223.2	456.6	364.0	356.1	365.6
1957	107.5	79.5	174.2	93.4	236.9	466.5	236.3	493.7	401.1	387.9	401.5
1958	116.4	82.3	177.4	99.0	257.9	527.8	251.6	571.7	448.6	436.4	449.9
1959	127.6	92.7	186.8	109.6	295.9	555.7	275.5	637.1	492.5	482.4	494.8
1960	138.7	102.7	178.9	119.5	337.3	583.0	285.2	710.2	533.4	516.7	534.2
1961	134.2	99.3	165.6	115.5	345.5	597.2	297.9	845.3	582.0	580.1	587.8
1962	109.1	99.1	193.1	104.9	363.4	626.0	324.9	913.4	623.3	630.4	632.2
1963	132.2	97.6	242.1	115.5	424.3	634.5	350.5	992.8	672.8	676.7	681.3
1964	152.7	98.4	273.0	125.7	467.1	661.2	370.8	1,073.3	720.3	727.4	730.3
1965	154.7	105.6	280.2	130.6	511.8	695.4	379.8	1,200.5	778.6	794.5	791.9
1966	144.7	113.3	304.1	130.8	522.7	690.4	403.8	1,241.7	801.5	841.0	822.2
1967	158.5	116.6	316.7	138.9	540.4	795.2	429.4	1,335.8	867.1	886.0	882.3
1968	168.6	119.6	413.9	147.3	597.6	815.8	476.6	1,395.7	921.8	927.5	933.6
1969	150.4	121.5	415.3	140.3	637.5	867.5	470.0	1,492.9	968.2	949.4	973.1
1970	162.1	124.7	423.0	147.3	689.9	878.4	471.0	1,598.2	1,013.5	991.8	1,018.1
1971	165.6	131.5	453.8	153.2	752.5	929.2	481.0	1,735.2	1,083.2	1,047.6	1,084.3
1972	137.9	134.6	462.1	142.8	773.4	944.2	481.2	1,722.9	1,088.3	1,043.3	1,086.7
1973	159.4	130.8	495.3	151.1	834.8	915.4	465.3	1,777.4	1,105.3	1,069.6	1,106.7
1974	152.6	144.3	534.8	156.2	888.2	825.3	499.5	1,778.9	1,111.6	1,097.6	1,119.6
1975	151.9	149.0	595.6	159.7	941.5	815.6	519.9	1,829.1	1,144.7	1,148.4	1,158.4
1976	147.7	156.6	640.8	163.0	1,031.9	826.5	540.0	1,894.7	1,195.8	1,165.5	1,199.7
1977	140.6	161.4	601.1	161.4	928.8	837.5	550.1	1,893.5	1,177.3	1,165.8	1,186.7
1978	106.4	160.5	548.7	144.5	914.3	854.7	565.4	1,895.1	1,184.5	1,153.0	1,188.0
1979	127.2	157.6	531.0	151.8	908.0	872.9	578.4	1,870.2	1,185.5	1,100.2	1,172.8
1980	143.6	165.5	569.6	164.1	916.9	869.1	596.8	1,920.3	1,208.4	1,130.2	1,198.1
1981	107.0	164.0	564.0	147.0	932.7	873.3	616.5	1,831.1	1,196.0	1,107.2	1,182.3
1982	130.0	149.7	555.9	149.5	898.0	885.1	604.3	1,846.3	1,189.7	1,083.9	1,170.8
1983	132.6	165.2	550.9	158.6	932.7	913.7	636.1	1,883.7	1,228.2	1,126.7	1,211.1

continued

Table A2 (continued)

Year	Crops	Livestock	Forestry, fishing	Primary	Food	Textiles	Other light	Heavy	Industry	Construction	Secondary
1984	142.6	174.5	595.5	169.0	966.5	922.2	677.2	1,863.0	1,249.2	1,137.9	1,229.4
1985	112.6	179.9	571.4	158.0	941.5	952.3	686.8	1,799.3	1,236.1	1,117.8	1,214.0
1986	142.8	181.1	583.9	172.3	945.4	972.9	699.8	1,813.3	1,251.1	1,131.3	1,228.7
1987	112.0	184.6	575.9	160.3	941.9	987.5	699.2	1,815.4	1,254.0	1,147.2	1,235.5
1988	99.2	183.2	586.9	154.1	957.1	986.6	735.0	1,820.6	1,273.5	1,153.5	1,251.3
1989	132.9	183.1	553.6	168.2	921.5	974.6	699.2	1,783.2	1,236.9	1,108.8	1,211.9
1990	110.2	176.6	482.3	153.0	828.2	938.6	608.8	1,602.8	1,118.3	978.4	1,088.4
1991	110.9	153.0	424.4	139.5	751.7	870.6	537.1	1,489.0	1,022.7	899.6	996.8
1992	91.9	120.7	192.5	108.6	464.8	510.6	332.8	772.2	584.2	483.8	560.4
1993	90.6	100.7	193.1	97.6	378.7	239.9	217.3	451.9	363.2	336.9	359.2
1994	81.6	88.1	210.8	87.4	374.0	274.6	264.3	430.2	383.1	355.4	378.9
1995	81.0	81.3	218.8	83.7	369.2	282.3	205.8	467.2	370.3	341.9	365.8

Sources: see online appendix. *Notes:* constant-price value-added volume indices, 1938 = 100; '—': not available. Commerce*: interwar index of trade, finance and insurance combined.

Table A3: Sectoral volume indices, Estonia 1919–95: services and total GDP

Year	Transport, comm.	Trade	Commerce*	Housing	Finance	Administration	Defence	Health	Education	Other	Tertiary	GDP
1919	30.3	—	32.1	46.2	—	—	470.2	—	—	93.9	81.7	56.0
1920	41.2	—	39.1	46.4	—	—	118.7	—	—	94.3	57.6	51.8
1921	50.1	—	48.3	47.2	—	—	100.0	—	—	96.0	62.2	59.8
1922	60.4	—	55.0	47.9	—	—	100.0	—	—	97.3	66.8	63.7
1923	63.9	—	67.9	48.6	—	—	100.0	—	—	98.1	72.9	69.9
1924	66.4	—	64.4	49.4	—	—	100.0	—	—	98.5	72.1	69.5
1925	65.8	—	70.3	50.1	—	—	100.0	—	—	98.6	75.2	74.9
1926	71.4	—	71.3	50.8	—	—	100.0	—	—	98.6	76.2	74.5
1927	70.3	—	73.6	51.6	—	—	100.0	—	—	98.5	77.0	75.0
1928	76.2	—	78.9	53.4	—	—	100.0	—	—	98.5	79.6	75.1
1929	83.3	—	78.5	55.2	—	—	100.0	—	—	98.5	81.1	77.5
1930	88.2	—	80.1	69.0	—	—	100.0	—	—	98.5	85.1	83.8
1931	86.2	—	71.4	86.2	—	—	100.0	—	—	98.8	83.2	80.0
1932	79.0	—	62.3	86.2	—	—	100.0	—	—	99.0	78.9	77.7
1933	71.6	—	66.1	82.8	—	—	100.0	—	—	99.3	79.5	80.8
1934	74.7	—	79.1	79.3	—	—	100.0	—	—	99.5	85.0	87.2
1935	84.9	—	83.0	86.2	—	—	100.0	—	—	99.7	88.4	87.4
1936	89.6	—	92.0	93.1	—	—	100.0	—	—	99.8	93.8	92.6
1937	95.4	—	102.2	96.6	—	—	100.0	—	—	99.8	100.0	101.2
1938	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
1939	112.0	100.3	100.3	100.6	100.3	101.5	100.0	107.1	91.1	99.6	102.1	102.6
1940	145.9	91.9	91.9	97.8	91.9	99.0	178.6	99.6	78.0	96.4	108.8	102.5

continued

Table A3 (continued)

Year	Transport, comm.	Trade	Commerce*	Housing	Finance	Administration	Defence	Health	Education	Other	Tertiary	GDP
1941	151.2	81.3	—	47.1	81.6	113.9	180.9	73.3	81.5	96.4	96.2	82.7
1942	130.8	79.2	—	69.9	79.8	128.7	183.1	82.5	85.1	96.4	103.2	79.1
1943	125.7	65.2	—	66.7	77.9	143.6	185.4	88.8	88.6	96.4	103.4	75.4
1944	101.5	52.6	—	63.4	9.0	93.4	165.9	88.8	63.1	42.3	87.1	57.8
1945	121.5	71.7	—	60.2	66.7	173.3	189.8	95.0	92.2	96.4	105.5	79.1
1946	183.1	81.3	—	70.9	71.1	176.3	193.1	129.2	104.6	131.2	115.5	87.7
1947	219.8	88.9	—	81.6	73.7	179.2	196.4	151.9	117.0	166.1	124.0	102.0
1948	211.1	96.6	—	92.3	76.2	182.2	199.6	176.7	125.8	200.9	130.8	112.5
1949	223.7	104.3	—	103.0	78.8	185.2	202.9	196.4	134.7	235.8	138.1	117.6
1950	252.5	114.9	—	113.7	83.5	188.2	206.1	199.3	159.0	270.6	147.0	125.4
1951	273.7	132.3	—	122.9	103.6	177.8	194.7	226.8	162.0	301.5	150.0	129.9
1952	301.3	147.0	—	125.4	113.8	167.4	183.3	237.6	166.5	316.9	150.2	134.3
1953	271.2	179.2	—	130.4	138.1	157.0	171.9	256.0	170.2	320.0	151.1	136.8
1954	297.1	209.5	—	131.3	144.9	146.6	160.6	280.5	184.3	353.9	154.0	141.0
1955	341.0	216.9	—	137.1	170.5	136.2	149.2	302.0	187.3	370.1	155.7	149.1
1956	372.2	233.4	—	136.3	194.0	133.7	146.5	315.7	201.4	388.6	159.9	162.1
1957	405.5	268.3	—	142.1	205.6	131.8	144.4	341.8	209.6	428.7	167.3	172.4
1958	431.9	284.0	—	152.2	209.6	130.0	142.4	347.9	217.0	492.7	173.3	185.2
1959	466.2	306.0	—	143.8	209.6	128.1	140.3	364.8	223.7	545.1	175.7	198.6
1960	517.8	341.9	—	148.8	196.9	126.3	138.3	383.2	229.6	634.9	183.1	212.6
1961	558.8	354.7	—	160.5	198.9	129.0	141.3	403.1	244.5	670.8	192.4	222.8
1962	617.8	368.5	—	161.4	197.2	131.7	144.3	412.3	253.4	704.7	198.3	227.2
1963	685.0	382.3	—	163.0	200.2	134.4	147.3	416.9	260.1	731.7	204.2	242.5
1964	777.1	402.5	—	167.2	210.7	137.2	150.2	432.2	270.5	817.3	214.0	259.1
1965	933.4	440.2	—	173.9	225.4	139.9	153.2	439.9	285.4	834.2	226.6	276.2
1966	981.3	485.2	—	175.6	247.7	146.1	160.0	446.0	296.5	956.8	238.3	285.5
1967	1,103.4	540.4	—	183.1	278.5	154.7	169.5	446.0	311.4	1,009.3	254.2	305.1
1968	1,169.2	601.9	—	186.5	304.9	163.4	179.0	475.2	321.0	1,022.4	268.0	322.7
1969	1,218.2	650.7	—	189.8	328.1	170.8	187.1	489.0	334.4	1,075.6	280.2	330.3
1970	1,303.4	704.0	—	189.0	355.0	175.8	192.6	495.1	341.9	1,125.7	290.9	344.9
1971	1,354.3	744.4	—	205.7	378.7	182.0	199.3	505.8	352.3	1,134.9	304.1	363.5
1972	1,379.9	786.7	—	215.7	414.7	184.4	202.0	516.5	364.1	1,169.6	314.7	362.7
1973	1,519.8	819.7	—	224.1	435.0	189.4	207.5	528.8	369.3	1,220.5	327.6	374.8
1974	1,683.6	851.0	—	230.8	452.7	195.6	214.3	538.0	376.8	1,289.9	341.4	384.4
1975	1,669.5	902.5	—	238.3	500.4	204.3	223.7	545.7	382.7	1,373.2	353.5	397.0
1976	1,730.6	961.3	—	237.5	529.0	212.9	233.2	553.3	390.9	1,394.8	365.0	409.5
1977	1,842.7	1,001.7	—	237.5	577.2	221.6	242.7	556.4	413.2	1,388.6	378.2	411.5
1978	1,901.7	1,035.7	—	242.5	601.1	230.2	252.2	567.1	426.6	1,475.0	390.9	408.2
1979	1,883.1	1,085.3	—	250.0	556.5	238.9	261.7	585.5	437.7	1,521.3	400.8	412.8
1980	1,880.2	1,159.8	—	258.4	592.7	247.6	271.2	597.8	450.4	1,549.8	414.8	428.1
1981	1,917.0	1,214.9	—	263.4	628.0	252.5	276.6	609.6	459.6	1,572.9	425.9	421.3

continued

Table A3 (continued)

Year	Transport, comm.	Trade	Commerce*	Housing	Finance	Administration	Defence	Health	Education	Other	Tertiary	GDP
1982	1,956.3	1,184.6	—	274.2	621.8	260.0	284.8	630.2	465.5	1,581.6	431.8	422.8
1983	2,024.2	1,212.2	—	287.6	634.4	257.5	282.1	644.9	468.8	1,595.2	439.8	436.8
1984	2,099.9	1,258.1	—	280.9	658.5	258.7	283.4	664.0	473.5	1,647.1	447.1	447.5
1985	2,098.9	1,326.1	—	283.4	688.1	230.2	252.2	687.6	478.7	1,876.5	448.7	440.2
1986	2,233.4	1,352.6	—	289.3	673.6	231.5	253.6	686.1	488.7	1,963.1	458.6	453.2
1987	2,527.3	1,379.2	—	286.8	719.3	231.5	253.6	695.0	501.2	1,819.6	468.9	452.3
1988	2,893.7	1,472.0	—	280.9	724.3	231.5	253.6	678.8	510.4	1,759.8	484.2	457.5
1989	3,162.3	1,564.8	—	280.1	817.3	212.9	233.2	730.3	519.0	1,868.6	499.4	463.4
1990	2,988.0	1,670.9	—	273.4	932.8	211.7	231.9	721.5	513.7	1,757.9	500.1	436.1
1991	2,903.8	951.7	—	255.0	581.9	160.4	220.3	731.7	502.4	1,697.2	423.6	386.5
1992	2,346.3	407.3	—	234.1	287.0	166.6	164.2	704.0	509.6	1,679.7	338.6	268.8
1993	2,573.3	583.3	—	231.6	420.9	179.4	111.3	689.4	514.8	1,676.5	346.3	233.3
1994	2,714.6	554.3	—	250.0	439.3	187.1	94.3	685.0	499.2	1,641.4	346.2	231.5
1995	1,486.7	661.8	—	269.2	499.9	183.5	86.4	532.0	578.0	1,842.1	327.4	220.7

Sources: see online appendix. *Notes:* constant-price value-added volume indices, 1938 = 100; '—': not available. Commerce*: interwar index of trade, finance and insurance combined.